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OGC Observations, Measurements and Samples

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Logical Model

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i. Keywords

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, Observations, Measurements, Samples, Monitoring

ii. Preface

The approved OGC Observations, Measurements and Samples (OMS) model is structured as three modeling layers which creates undue overhead for the implementation of domain models. Thus, a logical model layer compliant with the OMS model that provides all necessary associations is described in this document. Domain models created based on the OMS logical model will be compliant with OMS. In addition, concepts originating from the Environmental Monitoring Facilities Schema (EF) from the European INSPIRE Initiative are integrated as a valuable complement to the OMS Host class.

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iii. Security Considerations

No security considerations have been made for this Standard

iv. Submitting organizations

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

DataCove e.U.

Fraunhofer IOSB

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v. Submitters

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1. Scope

This standard defines a logical model and related abstract schema for observations, for features involved in the observation process, and for features involved in sampling when making observations. These provide models for the exchange of information describing observation acts and their results, both within and between different scientific and technical communities.

The OMS Logical Model has been simplified from the multi-tier Observations, Measurements and Samples standard, flattening the modelling hierarchy to provide all OMS concepts in one consistent model. While the complexity of the full OMS model enables a great deal of flexibility in integrating concepts from existing legacy systems within OMS based models, it also proved difficult to work with as all associations referenced the conceptual interfaces in place of classes. The OMS Logical Model ameliorates these issues, easing adoption of OMS in both existing and new systems.

Observations commonly involve sampling of an ultimate feature-of-interest. This document defines a common set of sample types according to their spatial, material (for ex situ observations) or statistical nature. The schema includes relationships between sample features (sub-sampling, derived samples).

This document concerns only externally visible interfaces and places no restriction on the underlying implementations other than what is needed to satisfy the interface specifications in the actual situation.

2. Conformance

This Standard defines an logical transposition of the OGC Observations, measurements and samples standard.

Conformance with this Standard shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site[1].

All requirements-classes and conformance-classes described in this document are owned by the Standard(s) identified.

1 www.opengeospatial.org/cite

3. References

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 19103:2015, *Geographic information — Conceptual schema language*

ISO 19107:2019, *Geographic information — Spatial schema*

ISO 19108:2002, *Geographic information — Temporal schema*

ISO 19109: 2015 *Geographic information — Rules for application schema*

ISO 19123:2005, *Geographic information — Schema for coverage geometry and functions*

ISO 19136-1:2020 Geographic information — Geography Markup Language (GML)

ISO 19156:2023, *Geographic information — Observation, measurements and samples*

4. Terms and Definitions

This document uses the terms defined in Policy Directive 49[2], which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word "shall" (not "must") is the verb form used to indicate a requirement to be strictly followed to conform to this Standard and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

For the purposes of this document, the following additional terms and definitions apply.

4.1
application schema
conceptual schema for data required by one or more applications

[SOURCE: ISO 19101-1:2014, 4.1.2]

[2] https://portal.ogc.org/public_ogc/directives/directives.php

4.2
coverage
feature that acts as a function to return values from its range for any direct position within its domain

[SOURCE: ISO/DIS 19123-1, 3.1.8]

4.3
data type
specification of a value domain with operations allowed on values in this domain

EXAMPLE Integer, Real, Boolean, String and Date.

Note 1 to entry: Data types include primitive predefined types and user-definable types.

[SOURCE: ISO 19103:2015, 4.14]

4.4
domain
well-defined set

Note 1 to entry: All elements within a domain (set) are of a given type.

[SOURCE: ISO 19109:2015, 4.8, modified — Original Note 1 to entry has been replaced with a new note to entry.]

4.5
domain feature
feature of a type defined within a particular application domain

Note 1 to entry: This can be contrasted with observations and sampling features, which are features of types defined for cross-domain purposes.

[SOURCE: ISO 19156:2023, 3.5]

4.6
ex situ
off-site
referring to the study, maintenance or conservation of a specimen or population away from its natural surroundings

Note 1 to entry: Opposite of *in situ* (on-site).

Note 2 to entry: an example of *ex situ* & direct is measuring a patient's temperature with a mercury thermometer in a blood-sample.

Note 3 to entry: an example of *ex situ* & remote is measuring a patient's temperature with an infra-red thermometer pointed at the blood sample.

[SOURCE: ISO 19156:2023, 3.6]

4.7
feature
abstraction of real world phenomena

Note 1 to entry: A feature can occur as a type or an instance. In this document, feature instance is meant unless otherwise specified.

[SOURCE: ISO 19101-1:2014, 4.1.11, modified — Note 1 to entry has been modified.]

4.8
feature-of-interest
subject of the observation

[SOURCE: ISO 19156:2023, 3.8]

4.9
feature type
class of features having common characteristics

[SOURCE: ISO 19156:2023, 3.9]

4.10
in situ
on-site
referring to the study, maintenance or conservation of a specimen or population without removing it from its natural surroundings

Note 1 to entry: Opposite of *ex situ* (off-site).

Note 2 to entry: an example of *in situ* & direct is measuring a patient's temperature with a mercury thermometer in the patient's rectum.

Note 3 to entry: an example of *in situ* & remote is measuring a patient's temperature with an infra-red thermometer at a distance.

[SOURCE: ISO 19156:2023, 3.10]

4.11
measure
<GML> value described using a numeric amount with a scale or using a scalar reference system

Note 1 to entry: When used as a noun, measure is a synonym for physical quantity.

[SOURCE: ISO 19136-1:2020, 3.1.41]

4.12
measurement
set of operations having the object of determining the value of a quantity

[SOURCE: ISO 19101-2:2018, 3.21]

4.13
observation
act carried out by an observer to determine the value of an observable property of an object (feature-of-interest) by using a procedure, with the value is provided as the result

[SOURCE: ISO 19156:2023, 3.13]

4.14
observation result
estimate of the value of a property determined through a known observation procedure

[SOURCE: ISO 19156:2011, 4.14]

4.15
observer
identifiable entity that can generate observations pertaining to an observable property by implementing a procedure

Note 1 to entry: An observer is an instance of a sensor, instrument, implementation of an algorithm or a being such as a person.

[SOURCE: ISO 19156:2023, 3.15]

4.16
positional accuracy
closeness of coordinate value to the true or accepted value in a specified reference system

Note 1 to entry: The phrase "absolute accuracy" is sometimes used for this concept to distinguish it from relative positional accuracy. Where the true coordinate value may not be perfectly known, accuracy is normally tested by comparison to available values that can best be accepted as true.

[SOURCE: ISO 19116:2025, 3.22]

4.17
procedure
specified way to carry out an activity or a process

[SOURCE: ISO 9000:2015, 3.4.5, modified — Note 1 to entry has been deleted.]

4.18
process
set of interrelated or interacting activities that use inputs to deliver an intended result

[SOURCE: ISO 9000:2015, 3.4.1, modified — Notes 1-6 have been deleted.]

4.19
property
facet or attribute of an object referenced by a name

EXAMPLE Abby's car has the colour red, where "colour red" is a property of the car.

Note 1 to entry: in some communities, the observed property is referred to as the measurand

[SOURCE: ISO 19143:2010, 4.21, modified — Example and note have been added to the entry.]

4.20
property type
characteristic of a feature type

EXAMPLE Cars (a feature type) all have a characteristic colour, where “colour” is a property type.

Note 1 to entry: The value for an instance of an observable property type can be estimated through an act of observation.

Note 2 to entry: In chemistry-related applications, the term “determinand” or “analyte” is often used.

[Source: ISO 19109:2005 Adapted]

4.21
proximate feature-of-interest
entity that is directly of interest in the act of observing

Note 1 to entry: This is a specialized form of the feature-of-interest

[SOURCE: ISO 19156:2023, 3.20]

4.22
range
<coverage> set of feature attribute values associated by a function, the coverage, with the elements of the domain of a coverage

Note 1 to entry: This is consistent with the more generic definition of range in ISO 19107:2019.

[SOURCE: ISO/DIS 19123-1, 3.1.47]

4.23
sample
object that is representative of a concept, real-world object or phenomenon

[SOURCE: ISO 19156:2023, 3.22]

4.24
sampler
device or entity (including humans) that is used by, or implements, a sampling procedure to create or transform one or more sample(s)

[SOURCE: ISO 19156:2023, 3.23]

4.25
sensor
element of a measuring system that is directly affected by a phenomenon, body, or substance carrying a quantity to be measured

[SOURCE: JCGM 200:2012, 3.8, modified — EXAMPLES and NOTE deleted.]

4.26
ultimate feature-of-interest
entity that is ultimately of interest in the act of observing

Note 1 to entry: This is a specialized form of the feature-of-interest.

[SOURCE: ISO 19156:2023, 3.25]

4.27
unit of measure
reference quantity chosen from a unit equivalence group

Note 1 to entry: In positioning services, the usual units of measurement are either angular units or linear units. Implementations of positioning services must clearly distinguish between SI units and non-SI units. When non-SI units are employed, it is required that their relation to SI units be specified.

[SOURCE: ISO 19116:2019, 3.29]

4.28
value
element of a type domain

Note 1 to entry: A value considers a possible state of an object within a class or type (domain).

Note 2 to entry: A data value is an instance of a datatype, a value without identity.

Note 3 to entry: A value can use one of a variety of scales including nominal, ordinal, ratio and interval, spatial and temporal. Primitive datatypes can be combined to form aggregate datatypes with aggregate values, including vectors, tensors and images.

[SOURCE: ISO/IEC 19501:2005, 0000_5, modified — Note 3 to entry has been added.]

5. Conventions

5.1 Abbreviated terms and acronyms

EF	INSPIRE Environmental Monitoring Facilities Model
FoI	Feature-of-Interest
GFM	General Feature Model

GML	Geography Markup Language
INSPIRE	Infrastructure for Spatial Information in Europe
O&M	Observations and Measurements
OMS	Observations, Measurements and Samples
OMS-L	OMS Logical Model (this current document)
OGC	Open Geospatial Consortium
STA	OGC SensorThings API
TSML	TimeSeries Modeling Language
UML	Unified Modeling Language
UoM	Unit of Measurement
XML	Extensible Markup Language
WIGOS	WMO Integrated Global Observing System
WMDR	WIGOS Metadata Representation
WMO	World Meteorological Organization

5.2 Schema language

The conceptual schema specified in this document is in accordance with the Unified Modelling Language (UML) ISO/IEC 19501, following the guidance of ISO 19103.

The UML in Abstract Core and Basic packages is conformant with the profile described in ISO 19136-1:2020, Annex E. Use of this restricted idiom supports direct transformation into a GML Application Schema. The stereotype «FeatureType» states that a class is an instance of the «metaclass» FeatureType (ISO 19109) [2], and therefore represents a feature type.

The prose explanation of the model uses the term “property” to refer to both class attributes and association roles. This is consistent with the General Feature Model described in ISO 19109. In the context of properties, the term “value” refers to either a literal (for attributes whose type is simple), or to an instance of the class providing the type of the attribute or target of the association. Within the explanation, the property names (property types) are sometimes used as natural language words where this assists in constructing a readable text.

5.3 Model element names

This Standard specifies a logical model for observations using terminology that is based on current practice in a variety of scientific and technical disciplines. It is designed to apply across disciplines, so the best or “most neutral” term has been used in naming the classes, attributes and associations provided. The terminology does not, however, correspond precisely with any single discipline.

5.4 Identifiers

The normative provisions in this specification are denoted by the URI

http://www.opengis.net/spec/{standard}/{m.n}

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

6. Applying this model (informative)

6.1 General

The current OGC OMS model consisting of three modeling layers creates undue overhead for the implementation of domain models. Thus, a logical model layer compliant with the OMS model that provides all necessary associations is described in this document.

OMS provides the relevant concepts for the structured description of observations, including the sampling structure, often essential for true understanding of the nature of the observations being provided. As data provisioning mechanisms are transitioning towards highly distributed linked approaches, the model structure and packaging have been significantly abstracted. This approach allows implementers to explicitly select the concepts to be supported based on their requirements, while clearly stating to which requirements and Conformance Classes their implementation complies. Both the Observation and Sample sections of this model have been structured using the following layering of packages:

- **Conceptual**: Within the Conceptual Model Packages, only Interfaces are provided. These models provide a very abstract view of the individual concepts they contain without reference to specific implementations. This approach allows for the inclusion of semantically aligned objects from external sources, that while having been created externally, do provide concepts sharing the same semantic meaning as the concepts from the Conceptual Models.
- **Abstract Core**: Within the Abstract Core Model Packages, only abstract featureTypes are provided following the semantic structure of the Conceptual model (i.e., realizing the interfaces provided by the Conceptual Model Packages). A consistent approach to metadata provision is introduced. All associations from the abstract featureTypes reference the conceptual interfaces for greater implementation flexibility. The Abstract Core Model Packages are designed for the creation of domain models. as required by individual communities. Due to their abstract definition, the classes from Abstract Core cannot be directly instantiated.
- **Basic**: Within the Basic Packages, simple concrete featureTypes (specializing the abstract ones from the Abstract Core model) are defined with some basic utility

attributes added for rapid out-of-the-box deployment. A few additional concepts pertaining to collections and potential observations are introduced at this level. This is the only level of OMS that can be directly instantiated.

While the layered approach described above allows for great flexibility (as required of a conceptual model), it also adds a serious burden on those tasked with specializing parts of OMS within their domain models. As part of the flexibility of the OMS layered approach, all associations across model layers target the interfaces from the conceptual model. In order to derive domain classes from the abstract classes provided by OMS, one must also include the interfaces for relevant context on associations. This is partially due to Enterprise Architect (the UML modeling tool used) not allowing display of inherited associations as can be done with inherited attributes. The resulting diagrams rapidly become unreadable, as shown in the simple example below.

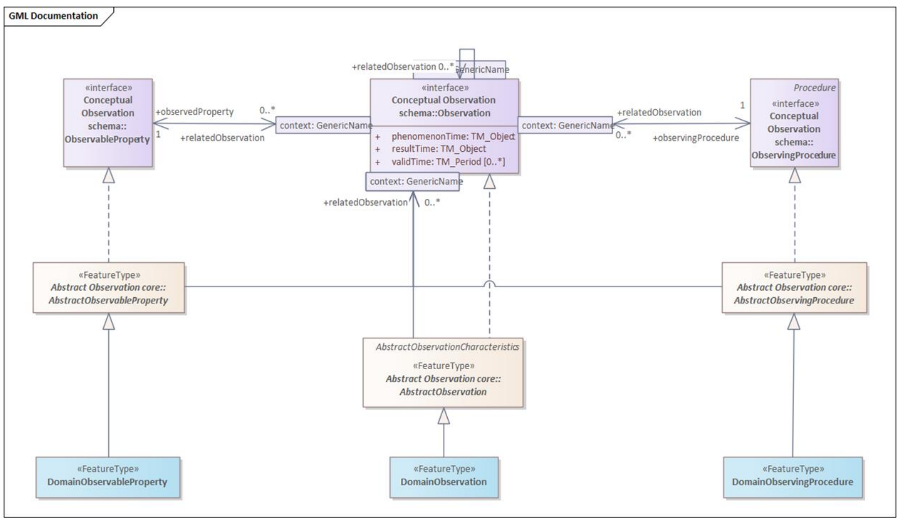

Figure 1: Modelling with full OMS Model

In this example, the Observation, ObservableProperty and ObservingProcedure types are specialized to domain types (e.g. because additional attributes are required). The top third of the diagram only serves the purpose of showing the associations between the interfaces. This adds additional unwanted overhead to models.

This issue was initially encountered during work done on updating the World Meteorological Organization (WMO) WIGOS Metadata Representation (WMDR) model to OMS. In this work it rapidly became apparent that the already complex WMDR models could not cope with the addition of the OMS interfaces together with the abstract classes. During the Time Series Modelling Language (TSML) update, contributors also encountered the same issue. As such, the current TSML work is based on the OMS Logical Model.

To overcome the above identified issue in the OMS Logical Model, all classes from the three layers of the OMS model are therefore provided as featureTypes, and all associations are available directly between these classes. As in the OMS model, this provides individual packages for Observation and Sampling. All these classes are aligned with the OMS model. However, not all classes are available at the conceptual level, e.g. the individual sample types SpatialSample, MaterialSample and StatisticalSample are not introduced until the OMS Basic Samples package. In these cases, the abstract class in the logical model had to be derived from the corresponding Basic Package.

6.2 Working with FeatureTypes

Within the OMS Logical model, the GML FeatureType Stereotype is used for all core classes. Thus, all classes of type FeatureType natively include the following basic attributes, in addition to geometry information:

- gml:id: A mandatory unique identifier for each feature instance, essential for referencing and linking between features. It must be unique within the scope of the XML document.
- gml:name: A label or name assigned to the feature, which may or may not be unique.
- gml:description: A text description of the feature.

Thus, these attributes are not modelled within this standard.

6.3 Creating Domain Classes

The OMS Logical model was created for the further derivation of bespoke domain models, tailoring the OMS structure towards specific requirements both by extending the existing classes as well as by defining additional classes and associations.

While the most straightforward approach is direct alignment, including one-to-one mapping of classes while maintaining class and attribute names, depending on the use case and target technology, modifications may be required. In some cases, classes will be renamed to better reflect domain concepts, while in other cases, classes may be merged if there is a clear one-to-one association between them.

This can be illustrated by the alignment of the OGC SensorThings API (STA) data model, shown in Figure 2, with OMS, shown in Figure 3 and Figure 4. At first glance, the models seem similar but different. However, with the following alignment, it becomes clear that STA fully conforms to OMS:

- Attribute renaming: OMS:Observation.parameter is renamed STA:Observation.properties
- Class renaming: OMS:Host is renamed STA:Thing; OMS:Observer is renamed STA:Sensor
- Class merge: OMS:ObservationCollection and OMS:ObservationCharacteristics have been merged to STA:Datastream

In this way, the OMS model can serve as the basis for a wide range of domain models, each tailored to the specific needs of their community, while maintaining structural interoperability.

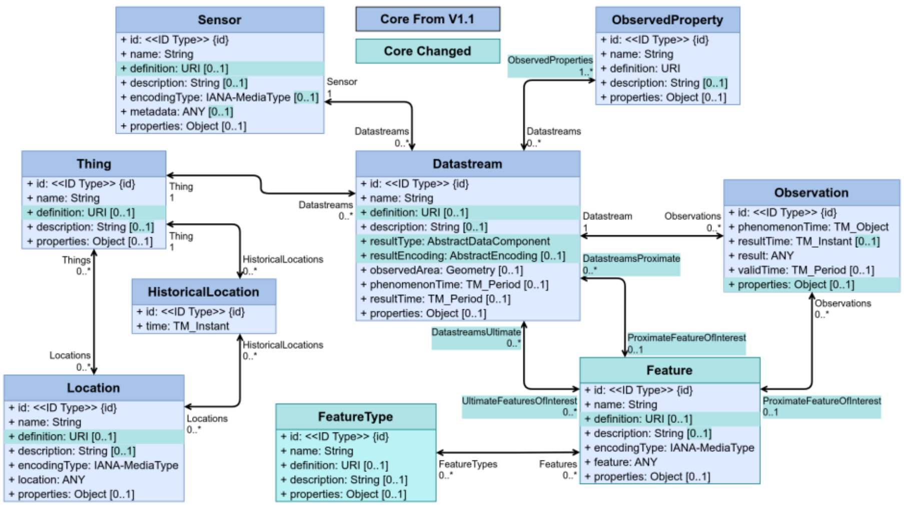

Figure 2: OGC SensorThings API data model

7. Fundamental characteristics of observations and samples (informative)

7.1 Observation Schema

7.1.1 General

The OMS Observation provides an efficient standardized method for providing information on properties of features. While the simplest approach for providing such feature information remains the definition of a simple attribute on this feature carrying this property, this approach is severely limited for providing additional relevant meta-information such as:

- The measurement methodology used in determining this property;
- The measurement device that provided the value;
- The facility at which this value was determined;
- The deployment time of the measurement device at the facility;
- Semantic details on the property being provided;
- Temporal support for properties that vary over time;
- Information on related observations.

In the process of defining specialized feature models for specific domains, it is valuable to reflect if this additional observational meta-information is of relevance for individual feature properties. When feature properties are static and observational meta-information is not relevant, a simple attribute will suffice; In situations where observational meta-information is relevant, e.g., environmental monitoring and reporting, utilization of the OMS Observation for provision of this feature property greatly eases professional data sharing.

7.1.2 Observation

An **Observation** is defined as an act carried out by an **Observer** to determine the value of an **ObservableProperty** of an object (**featureOfInterest)** by using an **ObservingProcedure**; the value is provided as the **result**. It is important to note that the terms ‘observation’, ‘interpretation’, ‘forecast’, ‘simulation’ correspond to this definition.

There are three different temporal properties on Observations

- **phenomenonTime**: The time for which the **result** applies to the characteristic of the **FeatureOfInterest** being observed. The **phenomenonTime** is often the time of interaction with a real-world feature either by a **SamplingProcedure** (time at which a **Sample** has been taken) or by an **ObservingProcedure**. If the result is the average of multiple samples taken at different times, then the phenomenonTime is the time interval over which these measurements were taken.
- **resultTime**: The instant of time when the **result** of the **Observation** became available. The **resultTime** typically corresponds to when the **Procedure** associated with the **Observation** was completed. For some observations this is identical to the **phenomenonTime**. However, there are important cases where they differ:
    - Where a measurement is made on a specimen in a laboratory, the **phenomenonTime** is the time the specimen was retrieved from its host, while the **resultTime** is the time the laboratory procedure was applied.
    - Where sensor observation results are post-processed, the **resultTime** is the post-processing time, while the **phenomenonTime** is the time of initial interaction with the world.
    - Simulations may be used to estimate the values for phenomena in the future or past. The **phenomenonTime** is the time that the result applies to, while the **resultTime** is the time that the simulation was executed.
- **validTime**: The time interval during which the **result** is assumed to be applicable for use. This attribute is commonly required in forecasting applications.

The **featureOfInterest** (FoI) association links to the feature that is the subject of the **Observation**. While in many cases, the **FoI** is a **Sample** (see section 7.2.2), any feature can serve as **FoI**. This object is either the real-world object whose properties are under observation, or it is an object used as a proxy for a real-world object that is not directly observable. Some examples of **FoI**s are:

- An instance of a feature modelled in a specific domain model (e.g., Borehole according to OGC GeoSciML).

- The bubble of air around the intake of an air quality monitoring station.
- An existing well being used for water quality measurements.

In some cases, the measurement process may be performed on an intermediary entity referred to as proximateFeatureOfInterest that acts as a proxy to the ultimate feature-of-interest that is being observed (measured, estimated or calculated). Thus, within the OMS model, there is a differentiation between:

- **proximateFeatureOfInterest**: The entity that is directly of interest in the act of observing.
- **ultimateFeatureOfInterest**: The entity that is ultimately of interest in the act of observing.

Some examples to illustrate the differentiation between proximate and ultimate FoI:

- To determine the concentrations of chemical compounds in a river, a sample is taken in a predefined location in the river. This sample is taken to a laboratory where the required chemical analysis is done. In this case, the river is the ultimateFeatureOfInterest, while the sample is the proximateFeatureOfInterest .
- Pertaining to documents and observations on the consistency thereof, for the Observation "This clause is inconsistent", the ultimateFeatureOfInterest is the entire document, while the proximateFeatureOfInterest is the specific clause being addressed.

Observational meta-information is provided by the following associations

- The **observedProperty** association links to the **ObservableProperty** that is the subject of the **Observation**. The **ObservableProperty** referenced by **observedProperty** should correspond to a characteristic associated with the **featureOfInterest**.
- The **observingProcedure** association links to the **ObservingProcedure** used by the **Observation** to determine the value of the **ObservableProperty** provided by the **result.** The **ObservingProcedure** referenced by **procedure** should be suitable for the associated **ObservableProperty**.
- The **observer** association links to the **Observer** that is involved in the creation of this **Observation**.
- The **host** association links to the **Host** that is involved in the creation of this **Observation**. At least one **Observer** or **Host** should be provided.

Finally, the **result** association links to the **result** of the **Observation**. The type of the result provided by the **result** association should be suitable for the associated **ObservableProperty**

The **Observation** will provide a unit of measure (UoM) if the result is measurable. If the UoM is not contained in the result, it is to be provided in the context of the **Observation**. The provision modality is to be defined by communities. The unit of measure should be suitable for the associated **ObservableProperty** and **ObservingProcedure**. In the case where the result of the Observation is a classification, for which no unit exists, the UoM can be declared as unitless

The **parameter** attribute provides a flexible extension point, allowing for the provision of name/value pairs further describing the **Observation**. **Parameter** should not be utilized to provide information already contained in the model by existing attributes or associations.

7.1.3 ObservableProperty

An **ObservableProperty** is defined as a quality (property, characteristic) of the **feature-of-interest** that can be observed. The height of a tree, the depth of a water body, or the temperature of a surface are examples of observable properties, while the value of a classic car is not (directly) observable but asserted.

The association **observer** links to an **Observer** capable of observing this **ObservableProperty**.

7.1.4 ObservingProcedure

An **ObservingProcedure** is defined as the description of steps performed in order to determine the value of an **observableProperty** by an **Observer**.

Depending on the complexity of the use case, the **ObservingProcedure** will be more or less explicitly described. Especially pertaining to historical data, there may be very little or no information available – what information is available should be provided;

The **ObservingProcedure** is a workflow, protocol, plan, algorithm, or computational method specifying how to make an observation. The **ObservingProcedure** is often referred to as the method, or the recipe that the observer (cook) follows to generate the **Observation**. Different **Observers** can follow the same (reusable) **ObservingProcedure** for the creation of different **Observations**.

The **ObservingProcedure** cannot describe a sensor instance, but it can describe the sensor type.

An instance of **ObservingProcedure** is a description of the process utilized by an **Observer**, this could be a chemical analysis method, a protocol for measuring an object, but could also be a checklist utilized by a human **Observer** during a biodiversity campaign. **ObservingProcedure** could further describe the algorithms behind simulators or models used to generate a result from other inputs.

The association **observer** links to an **Observer** capable of performing this **ObservingProcedure**.

7.1.5 Observer

An **Observer** is defined as an identifiable entity that can generate **Observations** pertaining to an **observableProperty** by implementing an **ObservingProcedure**.

The **Observer** is the entity instance of a sensor, instrument, implementation of an algorithm, or a being such as a person, not the entity type. It responds to a stimulus, e.g., a change in the environment, or input data composed from the results of prior **Observations**, and generates a **result**.

Pertaining to sensors, the **Observer** would reference the explicit sensor, while the **ObservingProcedure** would reference the methodology utilized by that sensor type. An **Observer** can be hosted by one or more **Host**. Different **Observer**s can follow the same (reusable) observing **ObservingProcedure** for the creation of different **Observations**. An **Observer** is closely linked with an **ObservableProperty** that it generates **result**s for;

Accelerometers, gyroscopes, barometers, magnetometers, and so forth are **Observer**s that are typically mounted on a modern smartphone (which acts as **Host**). Other examples of **Observer**s include the human eyes.

Contextual information to the **Observer** is provided via the following associations:

- The **observableProperty** association links to an **ObservableProperty** that this **Observer** can observe.
- The **observingProcedure** association links to an **ObservingProcedure** that this **Observer** can perform.
- The **deployment** association links to a **Deployment** to which this **Observer** is either physically or organizationally attached.

7.1.6 Host

A **Host** is defined as a grouping of **Observers** for a specific reason.

In many use cases, the **Host** is the environmental monitoring facility. The **Host** can be a platform that hosts a set of sensors. An alternative usage could pertain to a biodiversity survey campaign. In this scenario, the team performing the survey would be modelled as observers whereas the entire survey campaign can be represented as a **Host**.

The association **deployment** links to a **Deployment** at this **Host**.

The association **relatedHost** links to a **Host** the **Host** is related to.

7.1.7 Deployment

A **Deployment** is defined as information on the assignment of an **Observer** to a **Host**. Examples of **Deployment** are:

- Information regarding a sensor being attached to a pole.
- The monitoring facilities pertaining to an environmental monitoring network.
- The description of a ship cruise linking a research vessel with a marine network.
- The participation of a citizen in a citizen-science project involving crowd sensing.

The attribute **deploymentReason** provides a human readable description of the reason for the **Deployment**. The attribute **deploymentTime** provides the time that the **Deployment** pertains to.

Contextual information to the **Deployment** is provided via the following associations:

- The **observer** association links to the **Observer** associated with this **Deployment**.
- The **host** association links to the **Host** to which this **Deployment** pertains.

7.1.8 ObservationCharacteristics

An **ObservationCharacteristics** is defined as a set of common characteristics used for describing an **Observation** or a collection of Observations. **ObservationCharacteristics** has all of the attributes and associations as **Observation**, but none of the cardinality limitations (all attributes and associations have cardinality 0..*), making it a useful concept when describing observational matters.

7.1.9 ObservationCollection

An **ObservationCollection** is defined as a collection of similar **Observations**. The **ObservationCollection** is further described by the associated **ObservationCharacteristics** (linked via the **memberCharacteristics** association), that can carry all observational meta-information directly provided by **Observation**s.

The **collectionType** indicates how the linked **memberCharacteristics** impact the **Observation**s contained within the **ObservationCollection**. The following values are foreseen for **collectionType**:

- **homogeneous**: all observations contained are of a similar nature.
  If a property value is provided within the **ObservationCharacteristics**, this value applies to all **Observation**s contained in the **ObservationCollection**. The **Observations** need not contain attributes or associations supplied via the **ObservationCharacteristics**:
    - property not provided – values may be provided by the **Observation**s but is not provided at this level
    - property provided but with no content – no **Observation** within the collection provides this property

  - property = value – this value applies to all **Observation**s within the collection
  - property = value set/range – this value set/range applies to all **Observation**s within the collection
- **summarizing**: a wider "grab-bag" type of collection.
  If multiple values for a property are available in the contained **Observation**s, all values for this attribute (or the range of values contained in all **Observation**s) are provided in the **ObservationCharacteristics**. A property may also be empty in the **ObservationCharacteristics** – in this case any value can be provided for this attribute within the contained **Observation**s:
  - property not provided – values may be provided by the **Observation**s but a summary is not provided at this level;
  - property provided but with no content – no **Observation** within the collection provides this property;
  - property = value – this value applies to all **Observation**s within the collection;
  - property = value set/range – all **Observation**s provide a value within this set/range.

The **member** association links to all **Observation**(s) contained within the **ObservationCollection**.

The **relatedCollection** association links to an **ObservationCollection** related with this **ObservationCollection**.

7.1.10 ObservingCapability

An ObservingCapability is defined as information on Observation(s) that could potentially be provided. ObservingCapability is typically linked from a MonitoringFacility or Host, indicating what potential Observation(s) could be gathered from this facility.

EXAMPLE In order to explicitly describe the capabilities of an Environmental Monitoring Facility, information on what Observable Properties are being measured with which methodology is provided.

Commented [RA1]: EXAMPLEs cannot contain the verbal form "must". Please rephrase by using statement of fact.
Note that in accordance with the ISO House Style, the use of personal pronouns should also be avoided (in this case, "one").
A possible rephrasing of this sentence could be:
"..... information needs to be provided on what Obersable Properties..."

For example, in a national groundwater quantity monitoring network, depending on the equipment and the underlying observational strategies:

a) Some monitoring may have just one ObservingCapability:

1) ObservingCapability:

i) ultimateFeatureOfInterest: 'Hydrogeological Unit 121AS';

ii) proximateFeatureOfInterest:'xyz';

iii) procedure: 'Groundwater depth measurement by electronic probe';

iv) observedProperty: ‘GroundWaterDepth’.

b) Other monitoring may have several such ObservingCapabilities, for example:

1) ObservingCapability 1:

i) ultimateFeatureOfInterest: ‘Entite hydrogeologique 143AE05’;

ii) proximateFeatureOfInterest: ‘Calcaires du Muschelkalk de Lorraine à SERVIGNY-LES-RAVILLE’;

iii) procedure: ‘Groundwater depth measurement by electronic probe’;

iv) observedProperty: ‘GroundWaterDepth’.

2) ObservingCapability 2:

i) ultimateFeatureOfInterest: ‘Entite hydrogeologique 143AE05’;

ii) proximateFeatureOfInterest : ‘Calcaires du Muschelkalk de Lorraine à SERVIGNY-LES-RAVILLE’ ;

iii) procedure: ‘Digital recording teletransmitted’;

iv) observedProperty: ‘Water Temperature’.

3) ObservingCapability 3:

i) ultimateFeatureOfInterest: ‘Entite hydrogeologique 143AE05’;

ii) proximateFeatureOfInterest : ‘Calcaires du Muschelkalk de Lorraine à SERVIGNY-LES-RAVILLE’ ;

iii) procedure: ‘Digital recording teletransmitted’;

iv) observedProperty: ‘Water conductivity measured at 25°C’.

NOTE In the example above, URIs have been removed and only the labels provided for better readability.

7.2 Sample Schema

7.2.1 General

The OMS **Sample** allows for provision of information of **Sample**s, specifically for use as the **FoI** of an **Observation**. It enables provision of relevant meta-information on how **Samples** were derived and processed, including:

- The sampling act that resulted in the sample;
- The device used in obtaining the sample;

- The procedure followed in obtaining the sample;
- Individual preparation steps performed on the sample prior to measurement;
- The procedures followed in the individual preparation steps.

In addition, different types of **Sample**s have been specified:

- Spatial samples have a geospatial component;
- Material samples are physical tangible samples, often referred to as specimens.
- Statistical samples.

7.2.2 Sample

A **Sample** is defined as an object that is representative of a concept, real-world object or phenomenon. Typically, the **Sample** is a Feature which is intended to be representative of a **FeatureOfInterest** on which **Observation**s can be made. As such, it can carry a characteristic pertaining to the **ObservedProperty** being evaluated by the **Observation**.

The way the **Sample** is taken is typically guided by a sampling strategy. **Sample**s are often artefacts of an observational strategy and often have no significant function outside of their role in the observation process (although specimen preservation could be considered a specific activity per se).

For example, a profile typically samples a water- or atmospheric-column or a well samples the water in an aquifer or a tissue specimen samples a part of an organism. A statistical sample is often designed to be characteristic of an entire population. This is so that Observations can be made regarding the sample that provide a good estimate of the properties of the population.

The **sampleType** attribute can be used to indicate the type of **Sample** according to a community agreed typology.

Contextual information to the **Sample** is provided via the following associations:

- The **sampling** association links to the **Sampling** act the **Sample** is the result of.
- The **sampledFeature** association links to the feature the **Sample** is intended to be representative of.
- The **preparationStep** association links to the **PreparationStep**(s) applied to prepare the **Sample**.
- The **relatedSample** association links to a **Sample** the **Sample** is related to.

The **parameter** attribute provides a flexible extension point, allowing for the provision of name/value pairs further describing the **Sample**. **Parameter** should not be utilized to provide information already contained in the model by existing attributes or associations.

7.2.3 SpatialSample

A **SpatialSample** is defined as a geospatial **Sample**. When **Observation**(s) are made to estimate properties of a geospatial feature, in particular where the value of a property varies within the scope of the feature, a **SpatialSample** is used. Depending on accessibility and on the nature of the expected property variation, the **SpatialSample** can be extensive in one, two or three spatial dimensions.

Some common names for **SpatialSample** used in various application domains include Borehole, Flightline, Interval, Lidar Cloud, Map Horizon, Microscope Slide, Mine Level, Mine, Observation Well, Profile, Pulp, Quadrat, Scene, Section, ShipsTrack, Spot, Station, Swath, Trajectory, Traverse, etc.

The following attributes provide relevant spatial information:

- The **shape** attribute provides the Geometry of the **SpatialSample**.
- The **horizontalPositionalAccuracy** attribute provides information on the positional accuracy of the horizontal component of the Geometry of the **SpatialSample**.
- The **verticalPositionalAccuracy** attribute provides information on the positional accuracy of the vertical component of the Geometry of the **SpatialSample**.

7.2.4 MaterialSample

A **MaterialSample** is defined as a physical, tangible **Sample**. **MaterialSample**(s) that are curated and preserved are sometimes known as 'specimens'. **MaterialSample**(s) can be destroyed in connection with the observation act. A **MaterialSample** is a physical **Sample** of a **FeatureOfInterest**, obtained for **Observation**(s) normally carried out *ex situ*, sometimes in a laboratory.

The following attributes provide relevant information:

- The **size** attribute provides a physical extent of the **MaterialSample**. The size can be length, mass, volume, etc., as appropriate for the MaterialSample instance and its material type.
- The **storageLocation** attribute provides the location of a **MaterialSample**. The storageLocation can be a location such as a shelf in a warehouse or a drawer in a museum.
- The **sourceLocation** attribute provides the location from where the **MaterialSample** was obtained. The attribute **sourceLocation** of the **MaterialSample** can be unnecessary in case the related **Sampling** act

samplingLocation attribute is provided or when a relatedSample's location provides an unambiguous location.

7.2.5 StatisticalSample

A **StatisticalSample** is defined as a statistical subset of a feature-of-interest, defined for the purpose of creating **Observation**(s). **StatisticalSample**(s) usually apply to populations or other sets, of which certain subset can be of specific interest, e.g., the male or female subset of a population.

The following attributes provide relevant information:

- The **classification** attribute describes a criterion by which the subset was defined. The classification can be age, gender, etc., as appropriate for the set or population on which the subsetting is performed. A StatisticalClassification is provided for this purpose with the attributes:
  - The **concept** attribute defines the concept by which a **StatisticalClassification** is to be performed, e.g., age, gender, color, size etc..
  - The **classification** attribute provides the explicit classification classes pertaining to the classification concept described by the **StatisticalClassification**, e.g., Age Brackets: [0-10], [10-20]; Genders: Male, Female, Other; Color: Red, Green, Blue.

7.2.6 Sampling

A **Sampling** is defined as an act applying a **SamplingProcedure** to create or transform one or more **Sample**(s). Examples of **Sampling** are:

- Crushing a rock sample in a ball mill;
- Digging a pit through a soil sequence;
- Dividing a field site into quadrants;
- Drawing blood from a patient;
- Extracting water from an observation well;
- Extracting a sample from a defined environmental monitoring station;
- Registering an image of the landscape;
- Sieving a powder to separate the subset finer than 100-mesh;
- Selecting a subset of a population;
- Splitting a piece of drill-core to create two new samples;
- Taking a diamond-drill core from a rock outcrop.

The **samplingLocation** attribute can be used to provide location information pertaining to the **Sampling** act. The time attribute provides information on the time of the **Sampling**.

Contextual information for the **Sampling** act is provided via the following associations:

- The **sample** association links to the **Sample** generated by the **Sampling**;
- The **featureOfInterest** association links to the concept, real-world object or phenomenon (feature-of-interest) the **Sample**(s) of the **Sampling** represent;
- The **sampler** association links to the **Sampler** that performed the **Sampling**;
- The **samplingProcedure** association links to the **SamplingProcedure** used by the **Sampling**;
- The **relatedSampling** association links to related **Sampling**(s).

The **parameter** attribute provides a flexible extension point, enabling the provision of name/value pairs further describing the **Sampling** act. For example, when taking water samples, the sampling procedure specifies that an amount of time needs to pass to allow sediments to settle. The exact waiting time used in this Sampling can be stored in the parameter.

Parameter should not be utilized to provide information already contained in the model by existing attributes or associations.

7.2.7 Sampler

A **Sampler** is defined as a device or entity (including humans) that is used by, or implements, a **SamplingProcedure** to create or transform one or more **Sample**(s). Examples of **Sampler**s include:

- A ball mill, diamond drill, hammer;
- A hypodermic syringe and needle;
- An image sensor, a soil auger;
- A human being.

All the examples above can act as sampling devices (i.e., be **Samplers**). However, sometimes the distinction between the **Sampler** and the sensor (**Observer**) is not evident, as they are packaged as a unit. A **Sampler** need not be a physical device.

The **samplerType** attribute can be used to indicate the type of **Sampler** according to a community agreed typology.

Contextual information for the **Sampler** is provided via the following associations:

- The **sampling** association links to the **Sampling** act performed by the **Sampler**;

- The **implementedProcedure** association links to the **SamplingProcedure** implemented by the **Sampler**.

7.2.8 SamplingProcedure

A **SamplingProcedure** is defined as the description of steps performed by a **Sampler** in order to extract a **Sample** from its **sampledFeature** in the frame of a **Sampling**.

Contextual information to the **SamplingProcedure** is provided via the following associations:

- The **sampling** association can be used to link to the **Sampling** act(s) where this procedure has been used;
- The **sampler** association can be used to link to **Sampler**(s) that implement this procedure.

7.2.9 PreparationStep

A **PreparationStep** is defined as an individual step pertaining to a **PreparationProcedure**.

The **description** attribute can be used to provide a human readable description of the **PreparationStep**. The **time** attribute indicates the time the **PreparationStep** was performed.

Contextual information to the **PreparationStep** is provided via the following associations:

- The **processingDetails** association links to a **PreparationProcedure** step performed on the **Sample** the **PreparationStep** pertains to;
- The **preparedSample** association links to the **Sample** on which the **PreparationProcedure** is performed.

7.2.10 PreparationProcedure

A **PreparationProcedure** is defined as the description of preparation steps performed on a **Sample**.

The **samplePreparationStep** association can be used to link to the **PreparationStep** where this **PreparationProcedure** has been used.

7.2.11 SampleCollection

A **SampleCollection** is defined as a collection of **Sample**(s).

The **member** association links to all **Sample**(s) contained within the **SampleCollection**.

The **relatedCollection** association links to an **SampleCollection** related with this **SampleCollection**.

7.3 Monitoring Facilities Schema

7.3.1 General

The Environmental Monitoring Facilities Schema (EF) originates from the European INSPIRE Initiative[3] underpinned by the INSPIRE Directive (2007/2/EC[4]). When EF was originally defined, it was based on O&M, the predecessor to OMS. In this version of EF, it has been fully aligned with the OMS Logical Model. As the concepts covered in EF are applicable to a wide range of environmental monitoring undertakings, a core audience of the OMS standard, the framework defined in this standard brings this information to a wider audience.

The EF schema covers the following concepts:

- **Environmental Monitoring Facilities**: Installations/locations where environmental measurements take place, usually over time;
- **Environmental Monitoring Networks**: Describing administrative or organizational groupings of Environmental Monitoring Facilities;
- **Environmental Monitoring Programs** that define frameworks based on policy relevant documents defining the target of a collection of observations;
- **Environmental Monitoring Activities**: Circumscribing targeted activities in a coherent and concise timeframe, area and purpose.

As experience has shown that these concepts are useful beyond environmental sciences, the term Environmental has been dropped in the model.

Within the EF Schema sections of this document, liberal reuse of original INSPIRE texts are made.

7.3.2 MonitoringFacility

A georeferenced object directly collecting or processing data about objects whose properties (e.g. physical, chemical, biological or other aspects of environmental

[3] https://knowledge-base.inspire.ec.europa.eu/index_en

[4] https://eur-lex.europa.eu/eli/dir/2007/2/oj/eng

conditions) are repeatedly observed or measured. An environmental monitoring facility can also host other environmental monitoring facilities.

Laboratories are not MonitoringFacilities as the exact location of the laboratory does not add further information to the measurement. The methodology used in the laboratory should be provided with observational data.

The following attributes further characterize the **MonitoringFacility**:

- **mediaMonitored**: Monitored environmental medium;
- **legalBackground**: The legal context, in which the management and regulation of the **MonitoringFacility** is defined;
- **responsibleParty**: Responsible party for the **MonitoringFacility**;
- **onlineResource**: A link to an external document providing further information on the **MonitoringFacility**;
- **purpose**: Reason for which the **MonitoringFacility** was created;
- **reportedTo**: Information on the involvement of the AbstractMonitoringFeature in reporting;
- **representativePoint**: Representative location for the **MonitoringFacility**;
- **measurementRegime**: Regime of the measurement;
- **mobile**: Indicate whether the **MonitoringFacility** is mobile (repositionable) during the acquisition of the observation;
- **resultAcquisitionSource**: Source of result acquisition;
- **specializedMFType**: Categorization of **MonitoringFacility**(s) generally used by domain and in national settings.
  Examples: platform, site, station, sensor.

In addition, the following associations provide contextual information:

- **observingCapability**: A link pointing to the explicit capability of a **MonitoringFacility**. This provides a clean link between the observed property, the procedure used as well as the location of the measurement;
- **broader**: A link pointing to a broader **MonitoringFacility** (a higher level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **narrower**: A link pointing to narrower **MonitoringFacility**(s) (a lower level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **supersedes**: In a genealogy, the **MonitoringFacility**(s) that has(have) been deactivated/replaced by another one;
- **supersededBy**: In a genealogy, the newly active **MonitoringFacility**(s) that replaces(replace) the superseded one;
- **involvedIn**: **MonitoringActivity**(s) in which the **MonitoringFacility** is involved;

- **belongsTo**: A link pointing to the **MonitoringNetwork**(s) this **MonitoringFacility** pertains to. The association has additional properties as defined in the association class **NetworkFacility**;
- **operationalActivityPeriod**: Lifespan of the physical object (facility);
- **relatedTo**: Any Thematic Link to an Monitoring Facility. The association has additional properties as defined in the association class **AnyDomainLink**.

7.3.3 MonitoringNetwork

Administrative or organizational grouping of **MonitoringFacility**(s) managed the same way for a specific purpose, targeting a specific area. Each network respects common rules aiming at ensuring coherence of the observations, especially for purposes of **MonitoringFacility**(s), mandatory parameters selection, measurement methods and measurement regime.

The following attributes further characterize the **MonitoringNetwork**:

- **mediaMonitored**: Monitored environmental medium;
- **legalBackground**: The legal context, in which the management and regulation of the **MonitoringNetwork** is defined;
- **responsibleParty**: Responsible party for the **MonitoringNetwork**;
- **onlineResource**: A link to an external document providing further information on the **MonitoringNetwork**;
- **purpose**: Reason for which the **MonitoringNetwork** was created;
- **reportedTo**: Information on the involvement of the AbstractMonitoringFeature in reporting;
- **organizationLevel**: Level of legal organization the **MonitoringNetwork** is affiliated with.

In addition, the following associations provide contextual information:

- **observingCapability**: A link pointing to the explicit capability of a **MonitoringNetwork**. This provides a clean link between the observed property, the procedure used as well as the location of the measurement;
- **broader**: A link pointing to a broader **MonitoringNetwork** (a higher level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **narrower**: A link pointing to narrower **MonitoringNetwork**(s) (a lower level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **supersedes**: In a genealogy, the **MonitoringNetwork**(s) that has(have) been deactivated/replaced by another one;

- **supersededBy**: In a genealogy, the newly active **MonitoringNetwork**(s) that replaces(replace) the superseded one;
- **involvedIn**: **MonitoringActivity**(s) in which the **MonitoringNetwork** is involved;
- **contains**: A link pointing to the **MonitoringFacility**(s) included in this **MonitoringNetwork**. The association has additional properties as defined in the association class **NetworkFacility**.

7.3.4 MonitoringProgram

Framework based on policy relevant documents defining the target of a collection of observations and/or the deployment of **MonitoringFacility**(s) and **MonitoringNetwork**(s) on the field. Usually, an Monitoring Program has a long-term perspective over at least a few years.

The following attributes further characterize the **MonitoringProgram**:

- **mediaMonitored**: Monitored environmental medium;
- **legalBackground**: The legal context, in which the management and regulation of the **MonitoringProgram** is defined;
- **responsibleParty**: Responsible party for the **MonitoringProgram**;
- **onlineResource**: A link to an external document providing further information on the **MonitoringProgram**;
- **purpose**: Reason for which the **MonitoringProgram** was created.

In addition, the following associations provide contextual information:

- **observingCapability**: A link pointing to the explicit capability of a **MonitoringProgram**. This provides a clean link between the observed property, the procedure used as well as the location of the measurement;
- **broader**: A link pointing to a broader **MonitoringProgram** (a higher level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **narrower**: A link pointing to narrower **MonitoringProgram**(s) (a lower level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **supersedes**: In a genealogy, the **MonitoringProgram**(s) that has(have) been deactivated/replaced by another one;
- **supersededBy**: In a genealogy, the newly active **MonitoringProgram**(s) that replaces(replace) the superseded one;
- **triggers**: **MonitoringActivity**(s) triggered by the **MonitoringProgram**.

7.3.5 MonitoringActivity

Specific set of **MonitoringFacility**(s) and **MonitoringNetwork**(s) used for a given domain in a coherent and concise timeframe, area and purpose. Usually, the information collected is treated as a one-time step in a long-term monitoring Program. It is a concrete realization of a given **MonitoringProgram**.

For example, a vessel could be equipped with a collection of **MonitoringFacility**(s) for a given campaign (= **MonitoringActivity**) fulfilling a **MonitoringProgram** requirements. Then, after a given period this exact same vessel could be equipped with another set of **MonitoringFacilities** for another campaign fulfilling another **MonitoringProgram** requirements.

The following attributes further characterize the **MonitoringActivity**:

- **activityTime**: Lifespan of the MonitoringActivity;
- **activityConditions**: Textual description of the MonitoringActivity;
- **boundingBox**: Bounding box in which the MonitoringActivity takes place. Example: If a research vessel has several monitoring activities (MonitoringActivity) one wants to know where the vessel cruised for each of those (MonitoringActivity);
- **responsibleParty**: Responsible party for the MonitoringActivity;
- **onlineResource**: A link to an external document providing further information on the MonitoringActivity.

In addition, the following associations provide contextual information:

- **usesFacility**: Specific set of **MonitoringFacility**(s) involved in a **MonitoringActivity**;
- **usesNetwork**: Specific set of **MonitoringNetwork**(s) involved in a **MonitoringActivity**;
- **setUpFor**: **MonitoringProgram**(s) for which the **MonitoringActivity** is set up.

8. Logical Observation Schema (normative)

8.1 General

The Core Logical Observation schema is described as a class diagram in Figure 3.

In Figure 4 we see the additional classes ObservationCharacteristics and ObservationCollection.

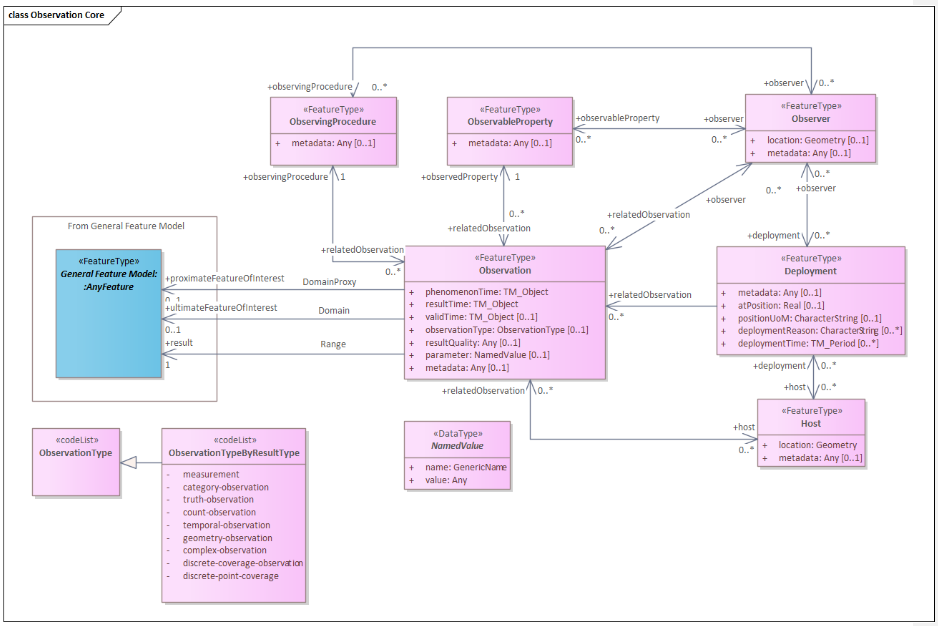

Figure 3: Core Logical Observation Schema

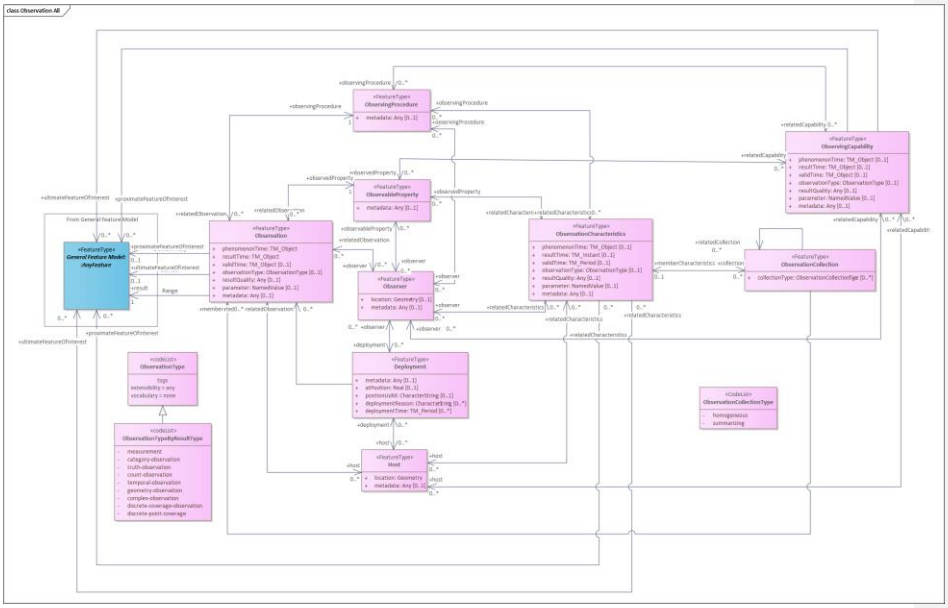

Figure 4: Logical ObservationCollection and ObservationCharacteristics Schema

Note: The OMS GenericDomainFeature class (ISO/DIS 19156 10.11) was not transposed into OMS-L, under the assumption that all actors who specialize their community models from OMS-L will define their own features as required.

8.2 Observation

Observation is the central class of the OMS Schema, tying together all relevant information required to describe an observation.

An **Observation** shall be defined as an act carried out by an **Observer** to determine the value of an **ObservableProperty** of an object (**featureOfInterest)** by using an **ObservingProcedure**; the value is provided as the **result**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Observation	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, CoreTypes conformance class
Dependency	ISO 19108:2002 Geographic information – Temporal schema, Application schemas for data transfer conformance class

Req 1	**/req/req-class-Observation/Observation-sem** An **Observation** shall be defined as an act carried out by an **Observer** to determine the value of an **ObservableProperty** of an object (**featureOfInterest)** by using an **ObservingProcedure**; the value is provided as the **result**.
Req 2	**/req/req-class-Observation/type-sem** Information providing further detail on the type of **Observations** being described. If information on the type of **Observation** is provided, the property **observationType:ObservationType** shall be used.
Req 3	**/req/req-class-Observation/observationType-sem** If information on the type of **Observation** is provided, the constraints defined in the referenced codelist shall be used.
Req 4	**/req/req-class-Observation/parameter-sem** Arbitrary event-specific parameter relevant to the **Observation**. If additional parameter information is provided, the property **parameter:NamedValue** shall be used.
Req 5	**/req/req-class-Observation/parameterName-card** The **name** attribute of a **parameter NamedValue** shall be unique within an **Observation**.
Req 6	**/req/req-class-Observation/resultQuality-sem** Information pertaining to the data quality of the **result**. If additional data quality information is provided, the property **resultQuality:Any** shall be used.
Req 7	**/req/req-class-Observation/phenomenonTime-sem** The time for which the **result** applies to the characteristic of the **FeatureOfInterest** being observed. If the **phenomenonTime** is described, this shall be provided by the attribute **phenomenonTime:TM_Object**
Req 8	**/req/req-class-Observation/phenomenonTime-card** An **Observation** shall have exactly 1 **phenomenonTime.**
Req 9	**/req/req-class-Observation/resultTime-sem** The instant of time when the **result** of the **Observation** became available.

	If the **resultTime** is described, this shall be provided by the attribute **resultTime:TM_Object**
Req 10	**/req/req-class-Observation/resultTime-card** An **Observation** shall have exactly 1 **resultTime**.
Req 11	**/req/req-class-Observation/resultTime-type** If the result time of the **Observation** is provided, the **resultTime** attribute shall be of type **TM_Instant**.
Req 12	**/req/req-class-Observation/validTime-sem** The time interval during which the **result** is assumed to be applicable for use. If **validTime(s)** is/are described, they shall be provided by the attribute **validTime:TM_Period**
Req 13	**/req/req-class-Observation/pFoI-sem** The entity that is directly of interest in the act of observing. If a reference to the entity being directly observed is provided, the association with the role **proximateFeatureOfInterest** shall be used. This association is a specialization of the **featureOfInterest** role.
Req 14	**/req/req-class-Observation/uFoI-sem** The entity that is ultimately of interest in the act of observing. If a reference to the entity ultimately being observed is provided, the association with the role **ultimateFeatureOfInterest** shall be used. This association is a specialization of the **featureOfInterest** role.
Req 15	**/req/req-class-Observation/featureOfInterest-con** At least one **proximateFeatureOfInterest** or **ultimateFeatureOfInterest** shall be given.
Req 16	**/req/req-class-Observation/observedProperty-sem** The **ObservableProperty** that is the subject of the **Observation**. If a reference to an **ObservableProperty** is provided, the association with the role **observedProperty** shall be used.
Req 17	**/req/req-class-Observation/observedProperty-card** An **Observation** shall have exactly 1 **observedProperty.**
Req 18	**/req/req-class-Observation/observedProperty-con**

	The **ObservableProperty** referenced by **observedProperty** shall correspond to a characteristic associated with the **featureOfInterest**
Req 19	**/req/req-class-Observation/result-sem** The **result** of the **Observation**. If a reference to a **result** is provided, the association **Range** with the role **result** shall be used.
Req 20	**/req/req-class-Observation/result-card** An **Observation** shall have exactly 1 **result**.
Req 21	**/req/req-class-Observation/result-con** The type of the result provided by the **result** association shall be suitable for the associated **ObservableProperty**
Req 22	**/req/req-class-Observation/observingProcedure-sem** The **ObservingProcedure** used by the **Observation** to determine the value of the **ObservableProperty** provided by the **result.** If a reference to an **ObservingProcedure** is provided, the association with the role **procedure** shall be used.
Req 23	**/req/req-class-Observation/observingProcedure-card** An **Observation** shall have exactly 1 **observingProcedure**.
Req 24	**/req/req-class-Observation/observingProcedure-con** The **ObservingProcedure** referenced by **procedure** shall be suitable for the associated **ObservableProperty**.
Req 25	**/req/req-class-Observation/observer-sem** An **Observer** that is involved in the creation of this **Observation**. If a reference to an **Observer** is provided, the association with the role **observer** shall be used.
Req 26	**/req/req-class-Observation/host-sem** A **Host** that is involved in the creation of this **Observation** If a reference to a **Host** is provided, the association with the role **host** shall be used.
Req 27	**/req/req-class-Observation/observerhost-con** At least one **Observer** or **Host** shall be provided
Req 28	**/req/req-class-Observation/uom**

	The **Observation** shall provide a unit of measure (UoM) if the result is measurable. If the UoM is not contained in the result, it shall be provided in the context of the **Observation**; the provision modality is to be defined by communities.
Req 29	**/req/req-class-Observation/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 30	**/req/req-class-Observation/metadata** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.
Rec 1	**/rec/req-class-Observation/parameter-procedure** **Parameter** should not be used instead of the procedure to describe the steps performed in order to determine the value of the ObservableProperty.
Rec 2	**/rec/req-class-Observation/parameter-redundant** **Parameter** should not be utilized to provide information already contained in the model by existing attributes or associations.
Rec 3	**/rec/req-class-Observation/uFoI** In the case where ultimate and proximate features-of-interest are the same object, the association should be provided using the **ultimateFeatureOfInterest** association role.
Rec 4	**/rec/req-class-Observation/phenomenonTimeResult-con** If the **observedProperty** of an **Observation** is ‘occurrence time’ then the **result** should be the same as the **phenomenonTime.**
Rec 5	**/rec/req-class-Observation/uom-con** The unit of measure should be suitable for the associated **ObservableProperty** and **ObservingProcedure**

8.3 ObservableProperty

An **ObservableProperty** shall be defined as a quality (property, characteristic) of the **feature-of-interest** that can be observed.

Requirements Class
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservableProperty

Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-ObservableProperty/ObservableProperty-sem** An **ObservableProperty** shall be defined as a quality (property, characteristic) of the **feature-of-interest** that can be observed.
Req 2	**/req/req-class-ObservableProperty/observer-sem** An **Observer** capable of observing this **ObservableProperty**. If a reference to the **Observer** is provided, the association with the role **observer** shall be used.
Req 3	**/req/req-class-ObservableProperty/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 4	**/req/req-class-ObservableProperty/metadata** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

8.4 ObservingProcedure

An **ObservingProcedure** shall be defined as the description of steps performed in order to determine the value of an **observableProperty** by an **Observer**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservingProcedure	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-ObservingProcedure/ObservingProcedure-sem** An **ObservingProcedure** shall be defined as the description of steps performed in order to determine the value of an **observableProperty** by an **Observer**.
Req 2	**/req/req-class-ObservingProcedure/observer-sem** An **Observer** capable of performing this **ObservingProcedure**.

	If a reference to an **Observer** is provided, the association with the role **observer** shall be used.
Req 3	**/req/req-class-ObservingProcedure/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 4	**/req/req-class-ObservingProcedure/metadata** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

8.5 Observer

An **Observer** shall be defined as an identifiable entity that can generate **Observations** pertaining to an **observableProperty** by implementing an **ObservingProcedure**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Observer	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-Observer/Observer-sem** An **Observer** shall be defined as an identifiable entity that can generate **Observations** pertaining to an **observableProperty** by implementing an **ObservingProcedure**.
Req 2	**/req/req-class-Observer/observableProperty-sem** An **ObservableProperty** that this **Observer** can observe. If a reference to **ObservableProperty**(s) is provided, the association with the role **observableProperty** shall be used.
Req 3	**/req/req-class-Observer/observingProcedure-sem** An **ObservingProcedure** that this **Observer** can perform. If a reference to **ObservingProcedure**(s) is provided, the association with the role **observingProcedure** shall be used.
Req 4	**/req/req-class-Observer/deployment-sem** A **Deployment** to which this **Observer** is either physically or organizationally attached.

	If a reference to **Deployment**(s) is provided, the association with the role **deployment** shall be used.
Req 5	**/req/req-class-Observer/location-sem** Location information pertaining to a feature**.** If location information is provided, the attribute **location:Geometry** shall be used.
Req 6	**/req/req-class-Observer/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 7	**/req/req-class-Observer/metadata** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

8.6 Host

A **Host** shall be defined as a grouping of **Observer**s for a specific reason.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Host	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-Host/Host-sem** A **Host** shall be defined as a grouping of **Observer**s for a specific reason.
Req 2	**/req/req-class-Host/deployment-sem** A **Deployment** at this **Host**. If a reference to a **Deployment** is provided, the association with the role **host** shall be used.
Req 3	**/req/req-class-Host/relatedHost-sem** A **Host** the **Host** is related to. If a reference to a related **Host** is provided, the association with role **relatedHost** shall be used. The **context:GenericName** qualifier of this

	association may be used to provide further information as to the nature of the relation.
Req 4	**/req/req-class-Host/location-sem** Location information pertaining to a feature**.** If location information is provided, the attribute **location:Geometry** shall be used.
Req 5	**/req/req-class-Host/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 6	**/req/req-class-Host/metadata** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

8.7 Deployment

A **Deployment** shall be defined as information on the assignment of an **Observer** to a **Host.**

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Deployment	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-Deployment/Deployment-sem** A **Deployment** shall be defined as information on the assignment of an **Observer** to a **Host.**
Req 2	**/req/req-class-Deployment/observer-sem** The **Observer** associated with this **Deployment**. If a reference to an **Observer** is provided, the association with the role **observer** shall be used.
Req 3	**/req/req-class-Deployment/host-sem** The **Host** to which this **Deployment** pertains.

	If a reference to a **Host** is provided, the association with the role **host** shall be used
Req 4	**/req/req-class-Deployment/deploymentReason-sem** A human readable description of the reason for the **Deployment**. If the reason for the **Deployment** is provided, the property **deploymentReason:CharacterString** shall be used.
Req 5	**/req/req-class-Deployment/deploymentTime-sem** The time that the **Deployment** pertains to. If the time of the **Deployment** is provided, property **deploymentTime:TM_Period** shall be used.
Req 6	**/req/req-class-Deployment/atPosition-sem** Position of the **Observer** relative to the **Host**. Up is positive, down is negative If the position of the **Deployment** is provided, the property **atPosition:Real** shall be used.
Req 7	**/req/req-class-Deployment/positionUoM-sem** Unit-of-Measure of the position of the **Observer** relative to the **Host**. If the position of the **Deployment** is provided, the property **positionUoM: CharacterString** shall be used.
Req 8	**/req/req-class-Deployment/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

8.8 ObservationCharacteristics

An **ObservationCharacteristics** shall be defined as a set of common characteristics used for describing an **Observation** or a collection of Observations.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCharacteristics	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-ObservationCharacteristics/ObservationCharacteristics-sem**

	An **ObservationCharacteristics** shall be defined as a set of common characteristics used for describing an **Observation** or a collection of Observations.
Req 2	**/req/req-class-ObservationCharacteristics/type-sem** Information providing further detail on the type of **Observations** being described by the **ObservationCharacteristics**. If information on the type of **Observation** is provided, the property **observationType:ObservationType** shall be used.
Req 3	**/req/req-class-ObservationCharacteristics/parameter-sem** Arbitrary event-specific parameter relevant to the **ObservationCharacteristics**. If additional parameter information is provided, the property **parameter:NamedValue** shall be used.
Req 4	**/req/req-class-ObservationCharacteristics/resultQuality-sem** Information pertaining to the data quality of the **result**. If additional data quality information is provided, the property **resultQuality:Any** shall be used.
Req 5	**/req/req-class-ObservationCharacteristics/phenomenonTime-sem** The time for which the **result** applies to the characteristic of the **FeatureOfInterest** being observed. If the **phenomenonTime** is described, this shall be provided by the attribute **phenomenonTime:TM_Object**
Req 6	**/req/req-class-ObservationCharacteristics/resultTime-sem** The instant of time when the **result** of the **Observation** became available. If the **resultTime** is described, this shall be provided by the attribute **resultTime:TM_Object**
Req 7	**/req/req-class-ObservationCharacteristics/validTime-sem** The time interval during which the **result** is assumed to be applicable for use. If **validTime(s)** is/are described they shall be provided by the attribute **validTime:TM_Period**
Req 8	**/req/req-class-ObservationCharacteristics/featureOfInterest-sem** The subject of the **Observation**.

	The reference(s) to **featureOfInterest(s)** shall be provided using the association **Domain** with the role **featureOfInterest**.
Req 9	**/req/req-class-ObservationCharacteristics/pFoI-sem** The entity that is directly of interest in the act of observing. If a reference to the entity being directly observed is provided, the association with the role **proximateFeatureOfInterest** shall be used. This association is a specialization of the **featureOfInterest** role.
Req 10	**/req/req-class-ObservationCharacteristics/uFoI-sem** The entity that is ultimately of interest in the act of observing. If a reference to the entity ultimately being observed is provided, the association with the role **ultimateFeatureOfInterest** shall be used. This association is a specialization of the **featureOfInterest** role.
Req 11	**/req/req-class-ObservationCharacteristics/collection-sem** An **ObservationCollection** that is described by these **ObservationCharacteristics**. If a reference to an **ObservationCollection** *from the* **ObservationCharacteristics** is provided, the association with the role **collection** shall be used.
Req 12	**/req/req-class-ObservationCharacteristics/observedProperty-sem** The **ObservableProperty** that is the subject of the **Observation**. If a reference to an **ObservableProperty** is provided, the association with the role **observedProperty** shall be used.
Req 13	**/req/req-class-ObservationCharacteristics/result-sem** The **result** of the **Observation**. If a reference to a **result** is provided, the association **Range** with the role **result** shall be used.
Req 14	**/req/req-class-ObservationCharacteristics/observingProcedure-sem** The **ObservingProcedure** used by the **Observation** to determine the value of the **ObservableProperty** provided by the **result.** If a reference to an **ObservingProcedure** is provided, the association with the role **procedure**shall be used.
Req 15	**/req/req-class-ObservationCharacteristics/observer-sem**

	An **Observer** that is involved in the creation of this **Observation**. If a reference to an **Observer** is provided, the association with the role **observer** shall be used.
Req 16	**/req/req-class-ObservationCharacteristics/host-sem** A **Host** that is involved in the creation of this **Observation** If a reference to a **Host** is provided, the association with the role **host** shall be used.
Req 17	**/req/req-class-ObservationCharacteristics/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 18	**/req/req-class-ObservationCharacteristics/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.
Rec 1	**/rec/req-class-ObservationCharacteristics/parameter-procedure** **Parameter** should not be used instead of the procedure to describe the steps performed in order to determine the value of the ObservableProperty.
Rec 2	**/rec/req-class-ObservationCharacteristics/parameter-redundant** **Parameter** should not be utilized to provide information already contained in the model by existing attributes or associations.
Rec 3	**/rec/req-class-ObservationCharacteristics/uFoI** In the case where ultimate and proximate features-of-interest are the same object, the association should be provided using the **ultimateFeatureOfInterest** association role.

8.9 ObservationCollection

An **ObservationCollection** shall be defined as a collection of similar **Observations.**

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCollection	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class

Req 1	**/req/req-class-ObservationCollection/ObservationCollection-sem** An **ObservationCollection** shall be defined as a collection of similar **Observations.**
Req 2	**/req/req-class-ObservationCollection/collectionType-sem** Information on the type of the **ObservationCollection.** If information on the collection type is provided, the attribute **collectionType:ObservationCollectionType** shall be used.
Req 3	**/req/req-class-ObservationCollection/collectionType-con** If the **collectionType** is provided, property values of the associated **Observation** and **ObservationCharacteristics** instances shall comply with the constraints defined for this **collectionType** value.
Req 4	**/req/req-class-ObservationCollection/member-sem** An **Observation** that is part of this **ObservationCollection**. If a reference to a member **Observation** is provided, the association with the role **member** shall be used.
Req 5	**/req/req-class-ObservationCollection/memberCharacteristics-sem** Information on **ObservationCharacteristics** of **Observations** contained within the **ObservationCollection**. If a reference to **ObservationCharacteristics** pertaining to the collection members is provided, the association with the role **memberCharacteristics** shall be used.
Req 6	**/req/req-class-ObservationCollection/relatedCollection-sem** A **ObservationCollection** the **ObservationCollection** is related to. If a reference to a related **ObservationCollection** is provided, the association with role **relatedCollection** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 7	**/req/req-class-ObservationCollection/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.

8.10 ObservingCapability

An **ObservingCapability** shall be defined as information on **Observation**(s) that could potentially be provided.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservingCapability	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-ObservingCapability/ObservingCapability-sem** An **ObservingCapability** shall be defined as information on **Observation**(s) that could potentially be provided.
Req 2	**/req/req-class-ObservingCapability/type-sem** Information providing further detail on the type of **Observations** being described by the **ObservingCapability**. If information on the type of **Observation** is provided, the property **observationType:ObservationType** shall be used.
Req 3	**/req/req-class-ObservingCapability/parameter-sem** Arbitrary event-specific parameter relevant to the **ObservingCapability**. If additional parameter information is provided, the property **parameter:NamedValue** shall be used.
Req 4	**/req/req-class-ObservingCapability/resultQuality-sem** Information pertaining to the data quality of the **result**. If additional data quality information is provided, the property **resultQuality:Any** shall be used.
Req 5	**/req/req-class-ObservingCapability/phenomenonTime-sem** The time for which the **result** applies to the characteristic of the **FeatureOfInterest** being observed. If the **phenomenonTime** is described, this shall be provided by the attribute **phenomenonTime:TM_Object**
Req 6	**/req/req-class-ObservingCapability/resultTime-sem** The instant of time when the **result** of the **Observation** became available. If the **resultTime** is described, this shall be provided by the attribute **resultTime:TM_Object**

Req 7	**/req/req-class-ObservingCapability/validTime-sem** The time interval during which the **result** is assumed to be applicable for use. If **validTime(s)** is/are described they shall be provided by the attribute **validTime:TM_Period**
Req 8	**/req/req-class-ObservingCapability/featureOfInterest-sem** The subject of the **Observation**. The reference(s) to **featureOfInterest(s)** shall be provided using the association **Domain** with the role **featureOfInterest**.
Req 9	**/req/req-class-ObservingCapability/pFoI-sem** The entity that is directly of interest in the act of observing. If a reference to the entity being directly observed is provided, the association with the role **proximateFeatureOfInterest** shall be used. This association is a specialization of the **featureOfInterest** role.
Req 10	**/req/req-class-ObservingCapability/uFoI-sem** The entity that is ultimately of interest in the act of observing. If a reference to the entity ultimately being observed is provided, the association with the role **ultimateFeatureOfInterest** shall be used. This association is a specialization of the **featureOfInterest** role.
Req 11	**/req/req-class-ObservingCapability/observedProperty-sem** The **ObservableProperty** that is the subject of the **Observation**. If a reference to an **ObservableProperty** is provided, the association with the role **observedProperty** shall be used.
Req 12	**/req/req-class-ObservingCapability/result-sem** The **result** of the **Observation**. If a reference to a **result** is provided, the association **Range** with the role **result** shall be used.
Req 13	**/req/req-class-ObservingCapability/observingProcedure-sem** The **ObservingProcedure** used by the **Observation** to determine the value of the **ObservableProperty** provided by the **result.** If a reference to an **ObservingProcedure** is provided, the association with the role **procedure**shall be used.

Req 14	**/req/req-class-ObservingCapability/observer-sem** An **Observer** that is involved in the creation of this **Observation**. If a reference to an **Observer** is provided, the association with the role **observer** shall be used.
Req 15	**/req/req-class-ObservingCapability/host-sem** A **Host** that is involved in the creation of this **Observation** If a reference to a **Host** is provided, the association with the role **host** shall be used.
Req 16	**/req/req-class-ObservingCapability/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 17	**/req/req-class-ObservingCapability/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.
Rec 1	**/rec/req-class-ObservingCapability/parameter-procedure** **Parameter** should not be used instead of the procedure to describe the steps performed in order to determine the value of the ObservableProperty.
Rec 2	**/rec/req-class-ObservingCapability/parameter-redundant** **Parameter** should not be utilized to provide information already contained in the model by existing attributes or associations.
Rec 3	**/rec/req-class-ObservingCapability/uFoI** In the case where ultimate and proximate features-of-interest are the same object, the association should be provided using the **ultimateFeatureOfInterest** association role.

8.11 NamedValue

The class **NamedValue** provides for a generic soft-typed parameter value.

Requirements Class
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-NamedValue

Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-NamedValue/NamedValue-sem** The class **NamedValue** provides for a generic soft-typed parameter value.
Req 2	**/req/req-class-NamedValue/name-sem** The attribute **name:GenericName** shall indicate the meaning of the named value.
Req 3	**/req/req-class-NamedValue/valu-sem** The attribute **value:Any** shall provide the value.

8.12 Codelist ObservationType

An empty extension-point for providing various classification schemes for **Observations**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationType	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-ObservationType/ObservationType-sem** An empty extension-point for providing various classification schemes for **Observations**. If **Observation** classification schemes are used in the implementing application schemas, a concrete realization shall be created for the application.

8.13 Codelist ObservationTypeByResultType

A codelist for classifying Observations by their result type

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationTypeByResultType	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-ObservationTypeByResultType/ObservationTypeByResultType-sem**

	The following entries shall be provided: • measurement: the result is of type **Measure**; • category-observation: the result is of type **ScopedName**; • truth-observation: result is a truth value; • count-observation: the result is of type **Integer**; • temporal-observation: the result is of type **TM_Object**; • geometry-observation: the result is of type **Geometry**; • complex-observation: the result is of type **Record**; • discrete-coverage-observation: result is a coverage that returns the same feature attribute values for every direct position within any single spatial object, temporal object, or spatiotemporal object in its domain; • discrete-point-coverage: result is a coverage that has a domain composed of points; • timeseries-observation: the result is a timeseries (a sequence of data values which are ordered in time).
Req 2	**/req/req-class-ObservationTypeByResultType/ObservationTypeByResultType-con** The following constraints shall be applied to the value of the result association of the **Observation** based on the codelist value used: • If the value "measurement" is used, the value of the result shall be of type **Measure**; • If the value "category-observation" is used the value of the result shall be of type **ScopedName**; • If the value "truth-observation" is used, the value of result shall be a truth value; • If the value "count-observation" is used, the value of the result shall be of type **Integer**;

	• If the value "temporal-observation" is used, the value of the result shall be of type **TM_Object**; • If the value "geometry-observation" is used, the value of the result shall be of type **Geometry**; • If the value "complex-observation" is used, the value of the result shall be of type **Record**.

8.14 Codelist ObservationCollectionType

A codelist for classifying ObservationCollections.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCollectionType	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-ObservationCollectionType/ObservationCollectionType-sem** The following entries shall be provided: • **homogeneous**: all observations contained are of a similar nature; • **summarizing**: a wider grab-bag type of collection.
Req 2	**/req/req-class-ObservationCollectionType/homogeneous-con** If **collectionType** in the **ObservationCollection** is specified as "**homogeneous**" from this Codelist, the following constraints apply to the associated **ObservationCharacteristics** and all **Observation** instances referenced via the member association. If a property value is provided within the **ObservationCharacteristics**, this value applies to all **Observations** contained in the **ObservationCollection**: • property not provided – values may be provided by the **Observations** but is not provided at this level property provided but with no content – no **Observation** within the collection provides this property • property = value – this value applies to all **Observations** within the collection

	property = value set/range – this value set/range applies to all **Observations** within the collection
Req 3	**/req/req-class-ObservationCollectionType/summarizing-con** If **collectionType** in the **ObservationCollection** is specified as "**summarizing**" from this Codelist, the following constraints apply to the associated **ObservationCharacteristics** and all **Observation** instances referenced via the member association. If multiple values for a property are available in the contained **Observations**, all values for this attribute (or the range of values contained in all **Observations**) are provided in the **ObservationCharacteristics**. A property may also be empty in the **ObservationCharacteristics** – in this case any value can be provided for this attribute within the contained Observations: • property not provided – values may be provided by the **Observations** but a summary is not provided at this level; • property provided but with no content – no **Observation** within the collection provides this property; • property = value – this value applies to all **Observations** within the collection; • property = value set/range – all **Observations** provide a value within this set/range.

9. Logical Sampling Schema (normative)

9.1 General

The Core Logical Sampling schema is described as a class diagram in Figure 5.

In Figure 6 the additional classes SampleCollection as well as the specialized Sample types StatisticalSample, MaterialSample and SpatialSample are presented.

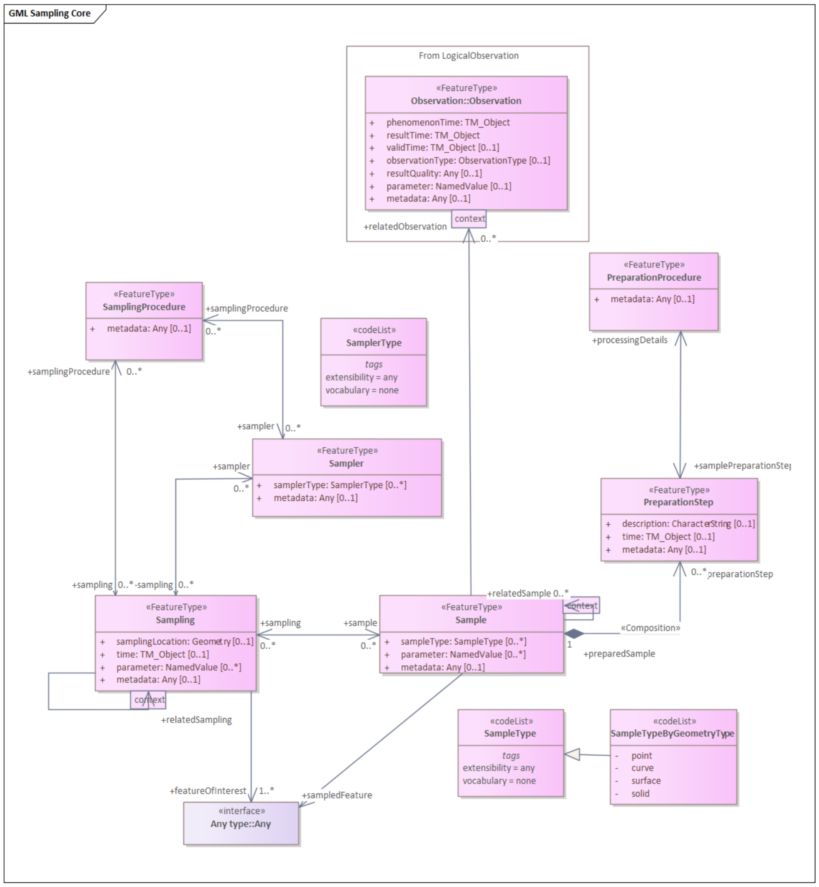

Figure 5: Core Sampling Schema

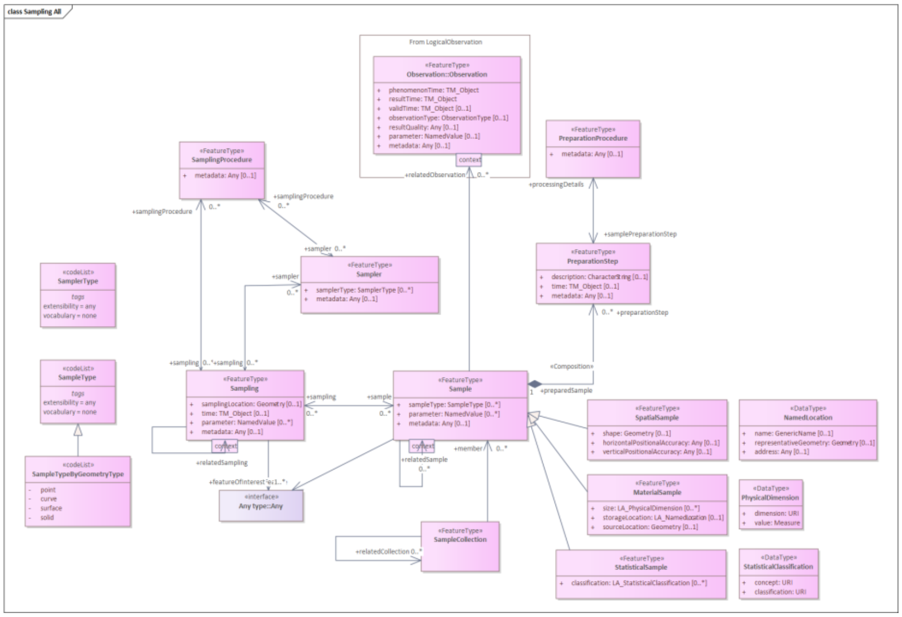

Figure 6: Full Sampling Schema

9.2 Sample

A **Sample** shall be defined as an object that is representative of a concept, real-world object or phenomenon.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sample	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-Sample/Sample-sem** A **Sample** shall be defined as an object that is representative of a concept, real-world object or phenomenon.
Req 2	**/req/req-class-Sample/sampling-sem** The **Sampling** the **Sample** is the result of. If **Sampling**(s) are described they shall be referred to using the association with the role **sampling**.

Req 3	**/req/req-class-Sample/preparationStep-sem** The **PreparationStep**(s) applied to prepare the **Sample**. If **PreparationSteps** are described, they shall be referred to using the association with the role **preparationStep**.
Req 4	**/req/req-class-Sample/sampledFeature-sem** The **sampledFeature** is the feature the **Sample** is intended to be representative of. References to the sampled feature shall be provided using the association with the role **sampledFeature**.
Req 5	**/req/req-class-Sample/sampleType-sem** The type of **Sample** according to a community agreed typology. If information on the type of **Sample** is provided, the attribute **sampleType:SampleType** shall be used.
Req 6	**/req/req-class-Sample/parameter-sem** Arbitrary event-specific parameter relevant to the **Sample**. If additional parameter information is provided, the property **parameter**:**NamedValue** shall be used.
Req 7	**/req/req-class-Sample/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.
Req 8	**/req/req-class-Sample/relatedSample-sem** A **Sample** the **Sample** is related to. If a reference to a related **Sample** is provided, the association with role **relatedSample** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 9	**/req/req-class-Sample/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.

9.3 SpatialSample

A **SpatialSample** shall be defined as a geospatial **Sample**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SpatialSample	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-SpatialSample/Sample-sem** A **Sample** shall be defined as an object that is representative of a concept, real-world object or phenomenon.
Req 2	**/req/req-class-SpatialSample/sampling-sem** The **Sampling** the **Sample** is the result of. If **Sampling**(s) are described they shall be referred to using the association with the role **sampling**.
Req 3	**/req/req-class-SpatialSample/preparationStep-sem** The **PreparationStep**(s) applied to prepare the **Sample**. If **PreparationSteps** are described, they shall be referred to using the association with the role **preparationStep**.
Req 4	**/req/req-class-SpatialSample/sampledFeature-sem** The **sampledFeature** is the feature the **Sample** is intended to be representative of. References to the sampled feature shall be provided using the association with the role **sampledFeature**.
Req 5	**/req/req-class-SpatialSample/sampleType-sem** The type of **Sample** according to a community agreed typology. If information on the type of **Sample** is provided, the attribute **sampleType:SampleType** shall be used.
Req 6	**/req/req-class-SpatialSample/parameter-sem** Arbitrary event-specific parameter relevant to the **Sample**. If additional parameter information is provided, the property **parameter**:**NamedValue** shall be used.
Req 7	**/req/req-class-SpatialSample/SpatialSample-sem** A **SpatialSample** shall be defined as a geospatial **Sample**.

Req 8	**/req/req-class-SpatialSample/shape-sem** The **shape** is the Geometry of the **SpatialSample**. If location information pertaining to the **SpatialSample** is provided, the attribute **shape:Geometry** shall be used.
Req 9	**/req/req-class-SpatialSample/horizontalPositionalAccuracy-sem** The positional accuracy of the horizontal component of the Geometry of the **SpatialSample**. If horizontal positional accuracy information pertaining to the **SpatialSample** is provided, the attribute **horizontalPositionalAccuracy:Any** shall be used.
Req 10	**/req/req-class-SpatialSample/verticalPositionalAccuracy-sem** The positional accuracy of the vertical component of the Geometry of the **SpatialSample**. If vertical positional accuracy information pertaining to the **SpatialSample** is provided, the attribute **verticalPositionalAccuracy:Any** shall be used.
Req 11	**/req/req-class-SpatialSample/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.
Req 12	**/req/req-class-SpatialSample/relatedSample-sem** A **Sample** the **Sample** is related to. If a reference to a related **Sample** is provided, the association with role **relatedSample** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 13	**/req/req-class-SpatialSample/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.

9.4 MaterialSample

A **MaterialSample** shall be defined as a physical, tangible **Sample**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MaterialSample	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MaterialSample/Sample-sem** A **Sample** shall be defined as an object that is representative of a concept, real-world object or phenomenon.
Req 2	**/req/req-class-MaterialSample/sampling-sem** The **Sampling** the **Sample** is the result of. If **Sampling**(s) is/are described, they shall be referred to using the association with the role **sampling**.
Req 3	**/req/req-class-MaterialSample/preparationStep-sem** The **PreparationStep**(s) applied to prepare the **Sample**. If **PreparationSteps** are described, they shall be referred to using the association with the role **preparationStep**.
Req 4	**/req/req-class-MaterialSample/sampledFeature-sem** The **sampledFeature** is the feature the **Sample** is intended to be representative of. References to the sampled feature shall be provided using the association with the role **sampledFeature**.
Req 5	**/req/req-class-MaterialSample/sampleType-sem** The type of **Sample** according to a community agreed typology. If information on the type of **Sample** is provided, the attribute **sampleType:SampleType** shall be used.
Req 6	**/req/req-class-MaterialSample/parameter-sem** Arbitrary event-specific parameter relevant to the **Sample**. If additional parameter information is provided, the property **parameter**:**NamedValue** shall be used.
Req 7	**/req/req-class-MaterialSample/MaterialSample-sem** A **MaterialSample** shall be defined as a physical, tangible **Sample**.
Req 8	**/req/req-class-MaterialSample/size-sem**

	The **size** describes a physical extent of the **MaterialSample**. If size information pertaining to the **MaterialSample** is provided, the attribute **size:PhysicalDimension** shall be used.
Req 9	**/req/req-class-MaterialSample/storageLocation-sem** The **storageLocation** is the location of a **MaterialSample**. If information pertaining to the storage location of the **MaterialSample** is provided, the attribute **storageLocation:NamedLocation** shall be used.
Req 10	**/req/req-class-MaterialSample/sourceLocation-sem** The **sourceLocation** is the location from where the **MaterialSample** was obtained. If information pertaining to the source location of the **MaterialSample** is provided, the attribute **sourceLocation:Geometry** shall be used.
Req 11	**/req/req-class-MaterialSample/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.
Req 12	**/req/req-class-MaterialSample/relatedSample-sem** A **Sample** the **Sample** is related to. If a reference to a related **Sample** is provided, the association with role **relatedSample** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 13	**/req/req-class-MaterialSample/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.

9.5 StatisticalSample

A **StatisticalSample** shall be defined as a statistical subset of a feature-of-interest, defined for the purpose of creating **Observation**(s).

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-StatisticalSample	
Target type	OMS Logical Model

Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-StatisticalSample/Sample-sem** A **Sample** shall be defined as an object that is representative of a concept, real-world object or phenomenon.
Req 2	**/req/req-class-StatisticalSample/sampling-sem** The **Sampling** the **Sample** is the result of. If **Sampling**(s) is/are described, they shall be referred to using the association with the role **sampling**.
Req 3	**/req/req-class-StatisticalSample/preparationStep-sem** The **PreparationStep**(s) applied to prepare the **Sample**. If **PreparationSteps** is/are described, they shall be referred to using the association with the role **preparationStep**.
Req 4	**/req/req-class-StatisticalSample/sampledFeature-sem** The **sampledFeature** is the feature the **Sample** is intended to be representative of. References to the sampled feature shall be provided using the association with the role **sampledFeature**.
Req 5	**/req/req-class-StatisticalSample/sampleType-sem** The type of **Sample** according to a community agreed typology. If information on the type of **Sample** is provided, the attribute **sampleType:SampleType** shall be used.
Req 6	**/req/req-class-StatisticalSample/parameter-sem** Arbitrary event-specific parameter relevant to the **Sample**. If additional parameter information is provided, the property **parameter**:**NamedValue** shall be used.
Req 7	**/req/req-class-StatisticalSample/StatisticalSample-sem** A **StatisticalSample** shall be defined as a statistical subset of a feature-of-interest, defined for the purpose of creating **Observation**(s).
Req 8	**/req/req-class-StatisticalSample/classification-sem** The **classification** describes a criterion by which the subset was defined.

	If information pertaining to the subsetting criteria by which a **StatisticalSample** is defined is provided, the attribute **classification:StatisticalClassification** shall be used.
Req 9	**/req/req-class-StatisticalSample/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.
Req 10	**/req/req-class-StatisticalSample/relatedSample-sem** A **Sample** the **Sample** is related to. If a reference to a related **Sample** is provided, the association with role **relatedSample** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 11	**/req/req-class-StatisticalSample/relatedObservation-sem** An **Observation** the object is related to. If a reference to a related **Observation** is provided, the association with role **relatedObservation** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.

9.6 Sampling

A **Sampling** shall be defined as an act applying a **SamplingProcedure** to create or transform one or more **Sample**(s).

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sampling	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-Sampling/Sampling-sem** A **Sampling** shall be defined as an act applying a **SamplingProcedure** to create or transform one or more **Sample**(s).
Req 2	**/req/req-class-Sampling/sample-sem** The **Sample** generated by the **Sampling.** If **Samples** are described, they shall be referred to using the association with the role **sample**.

Req 3	**/req/req-class-Sampling/featureOfInterest-sem** A **feature-of-interest** shall be defined as the concept, real-world object or phenomenon (feature-of-interest) the **Sample**(s) of the **Sampling** represent.
Req 4	**/req/req-class-Sampling/featureOfInterest-card** Reference to the feature-of-interest shall be done using the association with the role **featureOfInterest**.
Req 5	**/req/req-class-Sampling/sampler-sem** The **Sampler** that performed the **Sampling**. If **Sampler**(s) is/are described, they shall be referred to using the association with the role **sampler**.
Req 6	**/req/req-class-Sampling/samplingProcedure-sem** The **SamplingProcedure** used by the **Sampling.** If **SamplingProcedures** are described, they shall be referred to using the association with the role **samplingProcedure**.
Req 7	**/req/req-class-Sampling/samplingLocation-sem** If location information pertaining to the **Sampling** is provided, the attribute **samplingLocation:Geometry** shall be used.
Req 8	**/req/req-class-Sampling/time-sem** If information on the time of the **Sampling** is provided, the attribute **time:TM_Object** shall be used.
Req 9	**/req/req-class-Sampling/parameter-sem** Arbitrary event-specific parameter relevant to the **Sampling**. If additional parameter information is provided, the property **parameter**:**NamedValue** shall be used.
Req 10	**/req/req-class-Sampling/relatedSampling-sem** Related **Sampling**(s). If a reference to a related **Sampling** is provided, the association with role **relatedSampling** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 11	**/req/req-class-Sampling/metadata-sem**

	If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

9.7 Sampler

A **Sampler** shall be defined as a device or entity (including humans) that is used by, or implements, a **SamplingProcedure** to create or transform one or more **Sample**(s).

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sampler	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-Sampler/Sampler-sem** A **Sampler** shall be defined as a device or entity (including humans) that is used by, or implements, a **SamplingProcedure** to create or transform one or more **Sample**(s).
Req 2	**/req/req-class-Sampler/sampling-sem** The **Sampling** act performed by the **Sampler**. If **Sampling**(s) is/are described, they shall be referred to using the association with the role **sampling**.
Req 3	**/req/req-class-Sampler/implementedProcedure-sem** The **SamplingProcedure** implemented by the **Sampler**. If **SamplingProcedure**(s) is/are described, they shall be referred to using the association with the role **implementedProcedure**.
Req 4	**/req/req-class-Sampler/samplerType-sem** The type of **Sampler** according to a community agreed typology. If information on the type of **Sampler** is provided, the attribute **samplerType:SamplerType** shall be used.
Req 5	**/req/req-class-Sampler/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

9.8 SamplingProcedure

A **SamplingProcedure** shall be defined as the description of steps performed by a **Sampler** in order to extract a **Sample** from its **sampledFeature** in the frame of a **Sampling.**

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SamplingProcedure	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-SamplingProcedure/SamplingProcedure-sem** A **SamplingProcedure** shall be defined as the description of steps performed by a **Sampler** in order to extract a **Sample** from its **sampledFeature** in the frame of a **Sampling.**
Req 2	**/req/req-class-SamplingProcedure/sampling-sem** If the **SamplingProcedure** provides information on the **Sampling** where this procedure has been used, the association with the role **sampling** shall be used.
Req 3	**/req/req-class-SamplingProcedure/sampler-sem** If the **SamplingProcedure** provides information on the **Sampler** that implements this procedure, the association with the role **sampler** shall be used.
Req 4	**/req/req-class-SamplingProcedure/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

9.9 PreparationStep

A **PreparationStep** shall be defined as an individual step pertaining to a **PreparationProcedure**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PreparationStep	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-PreparationStep/PreparationStep-sem** A **PreparationStep** shall be defined as an individual step pertaining to a **PreparationProcedure**.
Req 2	**/req/req-class-PreparationStep/processingDetails-sem** A **PreparationProcedure** step performed on the **Sample** the **PreparationStep** pertains to.

	If **PreparationProcedure**(s) is/are described, they shall be referred to using the association with the role **processingDetails**.
Req 3	**/req/req-class-PreparationStep/preparedSample-sem** The **Sample** on which the **PreparationProcedure** is performed. The **Sample** shall be referred to using the association with the role **preparedSample**.
Req 4	**/req/req-class-PreparationStep/description-sem** Description of the **preparationStep**. If a description pertaining to the **preparationStep** is provided, the attribute **description:CharacterString** shall be used.
Req 5	**/req/req-class-PreparationStep/time-sem** Time of the **preparationStep**. If information on the time of the **preparationStep** of the **Sampling** is provided, the attribute **time:TM_Object** shall be used.
Req 6	**/req/req-class-PreparationStep/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

9.10 PreparationProcedure

A **PreparationProcedure** shall be defined as the description of preparation steps performed on a **Sample**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PreparationProcedure	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-PreparationProcedure/PreparationProcedure-sem** A **PreparationProcedure** shall be defined as the description of preparation steps performed on a **Sample**.
Req 2	**/req/req-class-PreparationProcedure/samplePreparationStep-sem** If the **PreparingProcedure** provides information on the **PreparationStep** where this procedure has been used, the association with the role **samplePreparationStep** shall be used.

Req 3	**/req/req-class-PreparationProcedure/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

9.11 SampleCollection

A **SampleCollection** shall be defined as a collection of **Sample**s.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleCollection	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-SampleCollection/SampleCollection-sem** A **SampleCollection** shall be defined as a collection of **Sample**s.
Req 2	**/req/req-class-SampleCollection/member-sem** A **Sample** that is part of this **SampleCollection**. If the **SampleCollection** has members, the association with the role **member** shall be used.
Req 3	**/req/req-class-SampleCollection/relatedCollection-sem** A **SampleCollection** the **SampleCollection** is related to. If a reference to a related **SampleCollection** is provided, the association with role **relatedCollection** shall be used. The **context:GenericName** qualifier of this association may be used to provide further information as to the nature of the relation.
Req 4	**/req/req-class-SampleCollection/metadata-sem** If descriptive metadata is provided, the association role **metadata** shall link to descriptive metadata as commonly understood by communities.

9.12 PhysicalDimension

A **PhysicalDimension** shall be defined as a dataType for the provision of various size quantities.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PhysicalDimension	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class

Req 1	**/req/req-class-PhysicalDimension/PhysicalDimension-sem** A **PhysicalDimension** shall be defined as a dataType for the provision of various size quantities.
Req 2	**/req/req-class-PhysicalDimension/dimension-sem** The name of the **PhysicalDimension** about which a **value** is provided. The identifier of the physical dimension shall be provided in the attribute **dimension:URI.**
Req 3	**/req/req-class-PhysicalDimension/value-sem** The **value** of the **PhysicalDimension**. The measure of the quantity being provided shall be provided in the attribute **value:Measure**

9.13 NamedLocation

A **NamedLocation** shall be defined as a location identified by its name, address, spatial geometry or a combination of any of these three.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-NamedLocation	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-NamedLocation/NamedLocation-sem** A **NamedLocation** shall be defined as a location identified by its name, address, spatial geometry or a combination of any of these three.
Req 2	**/req/req-class-NamedLocation/address-sem** An **address** used for identifying a **NamedLocation.** If **address** information is provided, the attribute **address:Any** shall be used.
Req 3	**/req/req-class-NamedLocation/name-sem** A **name** used for identifying a **NamedLocation.** If **name** information is provided, the attribute **name:GenericName** shall be used.
Req 4	**/req/req-class-NamedLocation/representativeGeometry-sem**

	A geometry used for providing a representative spatial location of a **NamedLocation.** If **geometry** is provided, the attribute **representativeGeometry:Geometry** shall be used.

9.14 StatisticalClassification

A **StatisticalClassification** shall be defined as a dataType for the provision of information on statistical classifications.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-StatisticalClassification	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-StatisticalClassification/StatisticalClassification-sem** A **StatisticalClassification** shall be defined as a dataType for the provision of information on statistical classifications.
Req 2	**/req/req-class-StatisticalClassification/concept-sem** The **concept** by which a **StatisticalClassification** is to be performed. The name of the concept by which the statistical classification is performed shall be provided in the attribute **concept:URI.**
Req 3	**/req/req-class-StatisticalClassification/classification-sem** The explicit **classification class** pertaining to the classification concept described by the **StatisticalClassification**. The classification class of the **StatisticalClassification** shall be provided in the attribute **classification:URI.**

9.15 Codelist SampleType

The code list SampleType can be specialized as required to more precisely define the semantics of sample types.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleType	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class

Req 1	**/req/req-class-SampleType/SampleType-sem** An empty extension-point for providing various classification schemes for **Samples**. If **Sample** classification schemes are used in the implementing application schemas, a concrete realization shall be created for the application.

9.16 Codelist SampleTypeByGeometryType

The code list SampleType can be specialized by geometry type

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleTypeByGeometryType	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-SampleTypeByGeometryType/SampleTypeByGeometryType-sem** The following entries shall be provided: • point: the provided geometry is of type **Point**. • curve: the provided geometry is of type **Curve**. • surface: the provided geometry is of type **Surface**. • solid: the provided geometry is of type **Solid**.
Req 2	**/req/req-class-SampleTypeByGeometryType/SampleTypeByGeometryType-con** The following constraints shall be applied to the value of the result association of the **Observation** based on the codelist value used: • If value “point” is used, the provided geometry shall be of type **Point**. • If value “curve” is used, the provided geometry shall be of type **Curve**. • If value “surface” is used, the provided geometry shall be of type **Surface**.

	• If value "solid" is used, the provided geometry shall be of type **Solid**.

9.17 Codelist SamplerType

The code list SamplerType can be specialized as required to more precisely define the semantics of sampler types.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SamplerType	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-SamplerType/SamplerType-sem** An empty extension-point for providing various classification schemes for **Samplers**. If **Sampler** classification schemes are used in the implementing application schemas, a concrete realization shall be created for the application.

10. Logical Monitoring Core (normative)

10.1 General

Within the European INSPIRE Environmental Monitoring Facilities specification, one finds the OMS Host concept fully fleshed out including relevant contextual information. As these have been proven useful in real world scenarios, they are being provided as an extension in the context of the OMS Logical model. As parts of the full INSPIRE model have a strong European focus, a simplified core model is derived, reduced to the most essential concepts. These are described in this section as the core model. In addition to the core model, the next section provides an extended version, including the full richness of the original INSPIRE model.

Please note that as all core classes are modelled as FeatureType, basic attributes such as identifier, name and geometry are already covered by this basic stereotype, they are not provided within this specification.

The core classes of this model, as shown in Figure 7, are:

- **MonitoringFacility**: A georeferenced object directly collecting or processing data about objects whose properties (e.g. physical, chemical, biological or other aspects of environmental conditions) are repeatedly observed or measured.
- **MonitoringNetwork**: Administrative or organizational grouping of MonitoringFacilities managed the same way for a specific purpose, targeting a specific area.
- **MonitoringActivity**: Specific set of MonitoringFacilities or MonitoringNetworks used for a given domain in a coherent and concise timeframe, area and purpose. Usually, the information collected is treated as one time step in a long term monitoring Program. It is a concrete realization of a given MonitoringProgram.
- **MonitoringProgram**: Framework based on policy relevant documents defining the target of a collection of observations and/or the deployment of MonitoringFacilities or MonitoringNetworks on the field. Usually, a Monitoring Program has a long-term perspective over at least a few years.

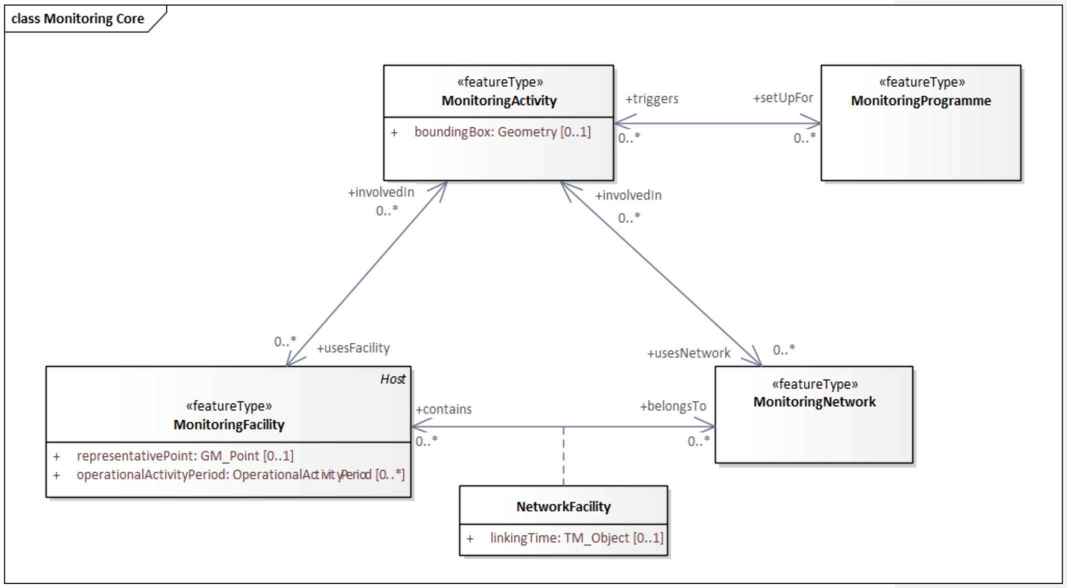

Figure 7: Core Monitoring Schema

In Figure 8 below, the Monitoring Schema is shown with OMS context, illustrating both how MonitoringFacility is derived from OMS:Host as well as how OMS:ObservingCapability is utilized by MonitoringFacility, MonitoringNetwork and MonitoringProgram.

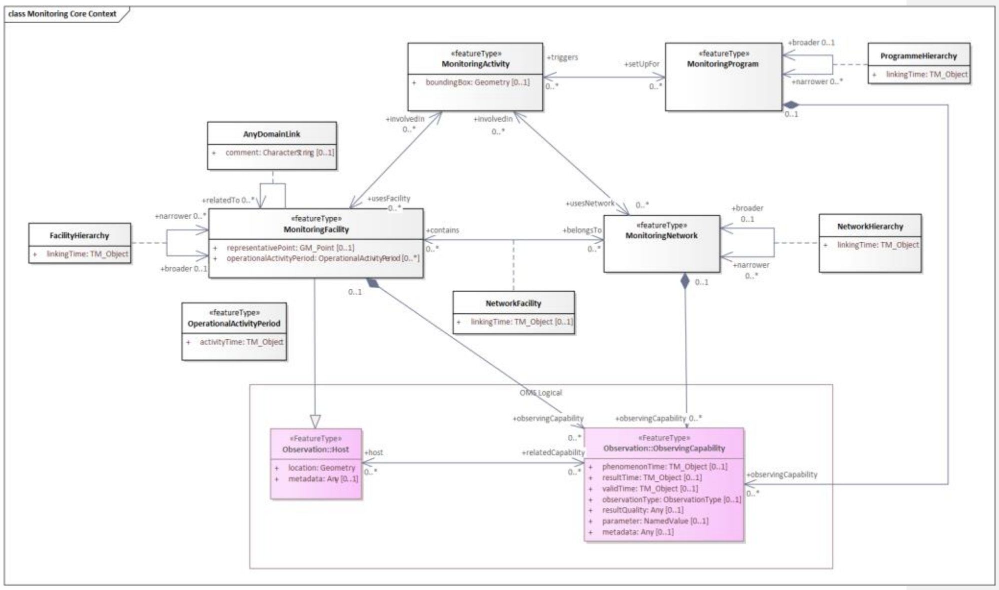

Figure 8: Monitoring Schema - OMS Context

10.2 MonitoringFacility

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacility	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacility/MonitoringFacility-sem** A georeferenced object directly collecting or processing data about objects whose properties (e.g. physical, chemical, biological or other aspects of environmental conditions) are repeatedly observed or measured. An environmental monitoring facility can also host other environmental monitoring facilities.
Req 2	**/req/req-class-MonitoringFacility/representativePoint** Representative location for the **MonitoringFacility**.
Req 3	**/req/req-class-MonitoringFacility/operationalActivityPeriod** Lifespan of the physical object (facility).

Req 4	**/req/req-class-MonitoringFacility/observingCapability** A link pointing to the explicit capability of a **MonitoringFacility**. This provides a clean link between the observed property, the procedure used as well as the location of the measurement.
Req 5	**/req/req-class-MonitoringFacility/broader** A link pointing to a broader **MonitoringFacility** (a higher level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**.
Req 6	**/req/req-class-MonitoringFacility/narrower** A link pointing to narrower **MonitoringFacility**(s) (a lower level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**.
Req 7	**/req/req-class-MonitoringFacility/belongsTo** A link pointing to the **MonitoringNetwork**(s) this **MonitoringFacility** pertains to. The association has additional properties as defined in the association class **NetworkFacility**.
Req 8	**/req/req-class-MonitoringFacility/relatedTo** Any Thematic Link to a **MonitoringFacility**. The association has additional properties as defined in the association class **AnyDomainLink**.
Req 9	**/req/req-class-MonitoringFacility/involvedIn** **MonitoringActivity**(s) in which the **MonitoringFacility** is involved.

10.3 MonitoringNetwork

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringNetwork/MonitoringNetwork-sem** Administrative or organizational grouping of **MonitoringFacilities** managed the same way for a specific purpose, targeting a specific area. Each network respects common rules aiming at ensuring coherence of the observations, especially for purposes of **MonitoringFacilities**, mandatory parameters selection, measurement methods and measurement regime.
Req 2	**/req/req-class-MonitoringNetwork/contains**

	A link pointing to the **MonitoringFacility**(s) included in this **MonitoringNetwork**. The association has additional properties as defined in the association class **NetworkFacility**.
Req 3	**/req/req-class-MonitoringNetwork/observingCapability** A link pointing to the explicit capability of a **MonitoringNetwork**. This provides a clean link between the observed property, the procedure used as well as the location of the measurement.
Req 4	**/req/req-class-MonitoringNetwork/broader** A link pointing to a broader **MonitoringNetwork** (a higher level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**.
Req 5	**/req/req-class-MonitoringNetwork/narrower** A link pointing to narrower **MonitoringNetwork**(s) (a lower level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**.
Req 6	**/req/req-class-MonitoringNetwork/involvedIn** **MonitoringActivity**(s) in which the **MonitoringNetwork** is involved.

10.4 MonitoringActivity

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivity	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringActivity/MonitoringActivity-sem** Specific set of **MonitoringFacility**(s) and/or **MonitoringNetwork**(s) used for a given domain in a coherent and concise timeframe, area and purpose. Usually, the information collected is treated as one time step in a long-term monitoring program. It is a concrete realisation of a given **MonitoringProgram**.
Req 2	**/req/req-class-MonitoringActivity/boundingBox** Bounding box in which the **MonitoringActivity** takes place.
Req 3	**/req/req-class-MonitoringActivity/usesFacility** Specific set of **MonitoringFacilities** involved in a **MonitoringActivity**.
Req 4	**/req/req-class-MonitoringActivity/usesNetwork** Specific set of **MonitoringNetworks** involved in a **MonitoringActivity**.

Req 5	**/req/req-class-MonitoringActivity/setUpFor** **MonitoringProgram**(s) for which the **MonitoringActivity** is set up.

10.5 MonitoringProgram

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgram	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgram/MonitoringProgram-sem** Framework based on policy relevant documents defining the target of a collection of observations and/or the deployment of **MonitoringFacility**(s) and/or **MonitoringNetwork**(s) on the field. Usually, a **MonitoringProgram** has a long-term perspective over at least a few years.
Req 2	**/req/req-class-MonitoringProgram/observingCapability** A link pointing to the explicit capability of a **MonitoringProgram**. This provides a clean link between the observed property, the procedure used as well as the location of the measurement.
Req 3	**/req/req-class-MonitoringProgram/broader** A link pointing to a broader **MonitoringProgram** (a higher level in a hierarchical structure). The association has additional properties as defined in the association class Hierarchy.
Req 4	**/req/req-class-MonitoringProgram/narrower** A link pointing to narrower **MonitoringProgram**(s) (a lower level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**.
Req 5	**/req/req-class-MonitoringProgram/triggers** A Link pointing to MonitoringActivity(s) triggered by the MonitoringProgram.

11. Logical Monitoring Extended (normative)

11.1 General

In this section an extended version of the Monitoring model is presented, providing additional attributes stemming from the original INSPIRE Environmental Monitoring Facilities specification. This has been done to enable reuse of these extended attributes as required within specific scenarios: Fine grained requirements classes enable mix and match of these extensions, as required to tailor custom solutions.

The core classes of this model, as shown in Figure 9, are:

- **MonitoringFacility**: A georeferenced object directly collecting or processing data about objects whose properties (e.g. physical, chemical, biological or other aspects of environmental conditions) are repeatedly observed or measured.
- **MonitoringNetwork**: Administrative or organizational grouping of MonitoringFacilities managed the same way for a specific purpose, targeting a specific area.
- **MonitoringActivity**: Specific set of MonitoringFacilities or MonitoringNetworks used for a given domain in a coherent and concise timeframe, area and purpose. Usually, the information collected is treated as one time step in a long-term monitoring program. It is a concrete realization of a given MonitoringProgram.
- **MonitoringProgram**: Framework based on policy relevant documents defining the target of a collection of observations and/or the deployment of MonitoringFacilities or MonitoringNetworks on the field. Usually, a Monitoring Program has a long-term perspective over at least a few years.

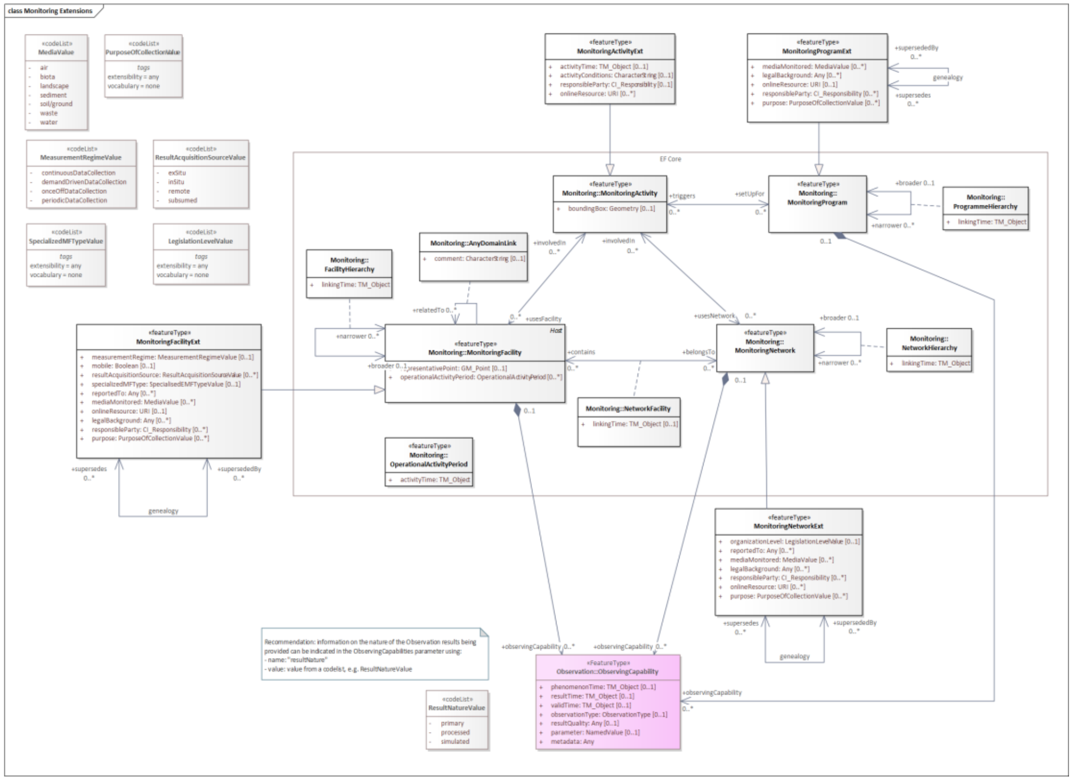

Figure 9: Extended Monitoring Schema

When utilizing the Monitoring Extensions in the creation of domain models, constraints can be useful to tailor the complexity of these types to specific use cases. In Figure 10, MyMonitoringFacility has been created based on MonitoringFacilityExt, adding a constraint mandating the provision of the mediaMonitored attribute.

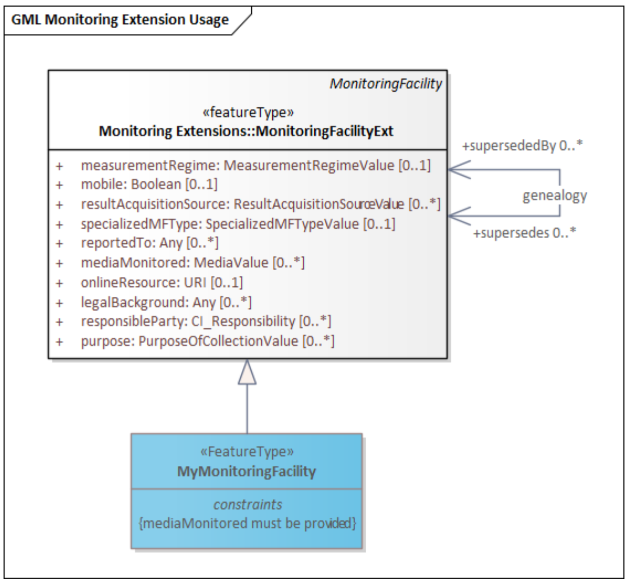

Figure 10: Using Extended Monitoring

11.2 MonitoringFacility Extended

11.2.1 reportedTo

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-reportedTo	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/reportedTo** Information on the involvement of the **MonitoringFacility** in reporting. If information on reporting activities is provided, the property **reportedTo:Any** shall be used.

11.2.2 mediaMonitored

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-mediaMonitored	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/mediaMonitored** Monitored environmental medium. If information on the environmental media being observed at the **MonitoringFacility** is provided, the property **mediaMonitored:MediaValue** shall be used.

11.2.3 legalBackground

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-legalBackground	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/legalBackground** The legal context, in which the management and regulation of the **MonitoringFacility** is defined. If information on the legal context of the **MonitoringFacility** is provided, the association with the role **legalBackground:Any** shall be used.

11.2.4 responsibleParty

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-responsibleParty	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/responsibleParty** Responsible party for the **MonitoringFacility**. If information on the responsible party of the **MonitoringFacility** is provided, the property **responsibleParty**:**CI_Responsibility** shall be used.

11.2.5 onlineResource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-onlineResource	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/onlineResource** A link to an external document providing further information on the **MonitoringFacility**. If a link to an external document providing further information on the **MonitoringFacility** is provided, the property **onlineResource:URI** shall be used.

11.2.6 purpose

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-purpose	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/purpose** Reason for which the **MonitoringFacility** was created. If information on the reason for which the **MonitoringFacility** was created is provided, the property **purpose:PurposeOfCollectionValue** shall be used.

11.2.7 measurementRegime

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-measurementRegime	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class

Req 1	**/req/req-class-MonitoringFacilityExt/measurementRegime** Regime of the measurement. If information on the regime of the measurement of the **MonitoringFacility** is provided, the property **measurementRegime:MeasurementRegimeValue** shall be used.

11.2.8 mobile

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-mobile	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/mobile** Indicate whether the **MonitoringFacility** is mobile (repositionable) during the acquisition of the observation. If indication whether the **MonitoringFacility** is mobile is provided, the property **mobile:Boolean** shall be used.

11.2.9 resultAcquisitionSource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-resultAcquisitionSource	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/resultAcquisitionSource** Source of result acquisition. If information on the source of result acquisition of the **MonitoringFacility** is provided, the property **resultAcquisitionSource:ResultAcquisitionSourceValue** shall be used.

11.2.10 specializedMFType

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-specializedMFType	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Dependency	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SpecializedMFTypeValue
Req 1	**/req/req-class-MonitoringFacilityExt/specializedMFType** Categorization of **MonitoringFacility**(s) generally used by domain and in national settings. If a categorization of the **MonitoringFacility**(s) is provided, the property **specializedMFType:specializedMFTypeValue** shall be used.

11.2.11 geometryRequired

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-geometryRequired	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/geometryRequired** **Geometry** and **representativePoint** shall not both be empty.

11.2.12 supersedes

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-supersedes	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/supersedes** In genealogy, the **MonitoringFacility**(s) that has(have) been deactivated/replaced by another one. If a reference to a previous version of the **MonitoringFacility** is provided, the association with the role **supersedes** shall be used.

11.2.13 supersededBy

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-supersededBy	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringFacilityExt/supersededBy** In genealogy, the newly active **MonitoringFacility**(s) that replaces(replace) the superseded one. If a reference to a subsequent version of the **MonitoringFacility** is provided, the association with the role **supersededBy** shall be used.

11.3 MonitoringNetwork Extended

11.3.1 mediaMonitored

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-mediaMonitored	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringNetworkExt/mediaMonitored** Monitored environmental medium. If information on the environmental media being observed by the **MonitoringNetwork** is provided, the property **mediaMonitored:MediaValue** shall be used.

11.3.2 legalBackground

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-legalBackground	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class

Req 1	**/req/req-class-MonitoringNetworkExt/legalBackground** The legal context, in which the management and regulation of the **MonitoringNetwork** is defined. If information on the legal context of the **MonitoringNetwork** is provided, the property **legalBackground:Any** shall be used.

11.3.3 responsibleParty

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-responsibleParty	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringNetworkExt/responsibleParty** Responsible party for the MonitoringNetwork. If information on the responsible party of the **MonitoringNetwork** is provided, the property **responsibleParty:responsibleParty** shall be used.

11.3.4 onlineResource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-onlineResource	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringNetworkExt/onlineResource** A link to an external document providing further information on the **MonitoringNetwork**. If a link to an external document providing further information on the **MonitoringNetwork** is provided, the property **onlineResource:URI** shall be used.

11.3.5 purpose

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-purpose	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringNetworkExt/purpose** Reason for which the **MonitoringNetwork** was created. If information on the reason for which the **MonitoringNetwork** was created is provided, the property **purpose:PurposeOfCollectionValue** shall be used.

11.3.6 organizationLevel

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-organizationLevel	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringNetworkExt/organizationLevel** Level of legal organization the **MonitoringNetwork** is affiliated with. If information on the level of legal organization the **MonitoringNetwork** is affiliated with is provided, the property **organizationLevel:LegislationLevelValue** shall be used.

11.3.7 supersedes

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-supersedes	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringNetworkExt/supersedes**

	In genealogy, the **MonitoringNetwork**(s) that has(have) been deactivated/replaced by another one. If a reference to a previous version of the **MonitoringNetwork** is provided, the association with the role **supersedes** shall be used.

11.3.8 supersededBy

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-supersededBy	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringNetworkExt/supersededBy** In a genealogy, the newly active **MonitoringNetwork**(s) that replaces(replace) the superseded one. If a reference to a subsequent version of the **MonitoringNetwork** is provided, the association with the role **supersededBy** shall be used.

11.4 MonitoringActivity Extended

11.4.1 activityTime

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-activityTime	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringActivityExt/activityTime** Lifespan of the **MonitoringActivity**. If information on the lifespan of the **MonitoringActivity** is provided, the property **activityTime:TM_Object** shall be used.

11.4.2 activityConditions

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-activityConditions	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringActivityExt/activityConditions** Textual description of the **MonitoringActivity**. If a textual description of the **MonitoringActivity** is provided, the property **activityConditions:CharacterString** shall be used.

11.4.3 responsibleParty

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-responsibleParty	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringActivityExt/responsibleParty** Responsible party for the **MonitoringActivity**. If information on the responsible party for the **MonitoringActivity** is provided, the property **responsibleParty:CI_Responsibility** shall be used.

11.4.4 onlineResource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-onlineResource	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringActivityExt/onlineResource** A link to an external document providing further information on the **MonitoringActivity**.

	If a link to an external document providing further information on the **MonitoringActivity** is provided, the property **onlineResource:URI** shall be used.

11.5 MonitoringProgram Extended

11.5.1 mediaMonitored

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-mediaMonitored	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt/mediaMonitored** Monitored environmental medium. If a reference to the environmental media being observed by the **MonitoringProgram** is provided, the association with the role **mediaMonitored** shall be used.

11.5.2 legalBackground

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-legalBackground	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt/legalBackground** The legal context, in which the management and regulation of the **MonitoringProgram** is defined. If information on the legal context of the **MonitoringProgram** is provided, the association with the role **legalBackground:Any** shall be used.

11.5.3 responsibleParty

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-responsibleParty	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt/responsibleParty** Responsible party for the **MonitoringProgram**. If information on the responsible party of the **MonitoringProgram** is provided, the property **responsibleParty**:**CI_Responsibility** shall be used.

11.5.4 onlineResource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-onlineResource	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt/onlineResource** A link to an external document providing further information on the **MonitoringProgram**. If a link to an external document providing further information on the **MonitoringProgram** is provided, the property **onlineResource:URI** shall be used.

11.5.5 purpose

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-purpose	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt/purpose**

	Reason for which the **MonitoringProgram** was created. If information on the reason for which the **MonitoringProgram** was created is provided, the property **purpose:PurposeOfCollectionValue** shall be used.

11.5.6 supersedes

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-supersedes	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt/supersedes** In genealogy, the **MonitoringProgram**(s)that has(have) been deactivated/replaced by another one. If a reference to a previous version of the **MonitoringProgram** is provided, the association with the role **supersedes** shall be used.

11.5.7 supersededBy

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-supersededBy	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt/supersededBy** In genealogy, the newly active **MonitoringProgram**(s) that replaces(replace) the superseded one. If a reference to a subsequent version of the **MonitoringProgram** is provided, the association with the role **supersededBy** shall be used.

11.6 Codelists

11.6.1 Codelist SpecializedMFTypeValue

An empty extension-point for providing various classification schemes for **MonitoringFacilities**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-SpecializedMFTypeValue	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt-SpecializedMFTypeValue/SpecializedMFTypeValue-sem** An empty extension-point for providing various classification schemes for **MonitoringFacilities**, generally used by domain and in national settings. EXAMPLES: platform, site, station, sensor, ...

11.6.2 Codelist PurposeOfCollectionValue

An empty extension-point for providing various classification schemes describing the reason for which **MonitoringFacilities**, **MonitoringNetworks** or **MonitoringPrograms** were created.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-PurposeOfCollectionValue	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt-PurposeOfCollectionValue/PurposeOfCollectionValue-sem** An empty extension-point for providing various classification schemes describing the reason for which **MonitoringFacilities**, **MonitoringNetworks** or **MonitoringPrograms** were created. EXAMPLES: European reporting obligation, international collaboration, long term monitoring of contaminated sites, ...

11.6.3 Codelist MediaValue

A codelist for classifying **MonitoringFacilities**, **MonitoringNetworks** or **MonitoringPrograms** by the media they monitor.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-MediaValue	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class

Req 1	**/req/req-class-MonitoringProgramExt-MediaValue/MediaValue-sem** A codelist for classifying **MonitoringFacilities**, **MonitoringNetworks** or **MonitoringPrograms** by the media they monitor. The following entries shall be included: • air • biota • landscape • sediment • soil/ground • waste • water

11.6.4 Codelist MeasurementRegimeValue

A codelist for classifying **MonitoringFacilities** by the measurement regime guiding their operation.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-MeasurementRegimeValue	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt-MeasurementRegimeValue/MeasurementRegimeValue-sem** A codelist for classifying **MonitoringFacilities** by the measurement regime guiding their operation. The following entries shall be provided: • **continuousDataCollection**: Data is collected on a continuous basis. There is usually no end date, as further data is collected. • **demandDrivenDataCollection**: Data is collected on demand. • **onceOffDataCollection**: Data is collected only once in this configuration. No further observations in this configuration can be expected. • **periodicDataCollection**: Data is collected at regular intervals. No information is available as to the data collection interval.

11.6.5 Codelist ResultAcquisitionSourceValue

A codelist for classifying **MonitoringFacilities** by their result acquisition source.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-ResultAcquisitionSourceValue	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt-ResultAcquisitionSourceValue/ResultAcquisitionSourceValue-sem** A codelist for classifying **MonitoringFacilities** by their result acquisition source. The following entries shall be provided: • **exSitu**: The FeatureOfInterest is a specimen taken from the ultimate FeatureOfInterest (i.e. the sampledFeature). • **inSitu**: The FeatureOfInterest is a sampling feature which is co-located with the ultimate FeatureOfInterest (i.e. the sampledFeature). • **remote**: The FeatureOfInterest is a sampling feature which is also the ultimate FeatureOfInterest (i.e. the sampledFeature). • **subsumed**: The value is inherited from children.

11.6.6 Codelist ResultNatureValue

A codelist for classifying **MonitoringFacilities**, **MonitoringNetworks** or **MonitoringPrograms** by the nature of their results.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-ResultNatureValue	
Target type	OMS Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	**/req/req-class-MonitoringProgramExt-ResultNatureValue/ResultNatureValue-sem** A codelist for classifying **MonitoringFacilities**, **MonitoringNetworks** or **MonitoringPrograms** by the nature of their results. The following entries shall be provided:

	• **primary**: The result provided with the observation is the direct result of an estimate of a property on the featureOfInterest. No further processing was performed. Processing may have taken place, but only in the sense of the measurement methodology itself, i.e. converting the millivolt returned from the sensor to the concentration of a substance. • **processed**: The result provided, while usually based on primary measurements, has been substantially processed. This processing can be of diverse natures, in some situations complex aggregates are provided, in other situations, the existing values are interpolated to a continuum. • **simulated**: The result provided, while usually based on primary measurements, is based on an interpretation model, and provides a simulation of past or future states of the media being analyzed. In this case, the existing values are usually extrapolated into the past or future.

Annex

Observation

Annex A: Conformance Class Test Suite (Normative)

Conformance class: Observation

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Observation		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Observation**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Observation/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservableProperty

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservableProperty		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservableProperty**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservableProperty/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservingProcedure

Conformance Class
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservingProcedure

Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservingProcedure**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservingProcedure/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Observer

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Observer		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Observer**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Observer/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Host

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Host		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Host**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Host/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Deployment

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Deployment		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Deployment**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Deployment/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationCharacteristics

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCharacteristics		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCharacteristics**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCharacteristics/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationCollection

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCollection	
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCollection**

Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCollection/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservingCapability

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservingCapability		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservingCapability**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservingCapability/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: NamedValue

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-NamedValue		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-NamedValue**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-NamedValue/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationType		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationType**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationType/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationTypeByResultType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationTypeByResultType		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationTypeByResultType**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationTypeByResultType/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationCollectionType

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCollectionType	
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCollectionType**

Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCollectionType/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Sampling

Conformance class: Sample

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sample		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sample**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sample/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SpatialSample

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SpatialSample		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SpatialSample**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SpatialSample/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.

	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MaterialSample

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MaterialSample		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MaterialSample**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MaterialSample/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: StatisticalSample

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-StatisticalSample		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-StatisticalSample**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-StatisticalSample/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Sampling

Conformance Class

http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sampling		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sampling**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sampling/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Sampler

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sampler		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sampler**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sampler/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SamplingProcedure

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SamplingProcedure		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SamplingProcedure**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SamplingProcedure/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.

	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: PreparationStep

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PreparationStep		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PreparationStep**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PreparationStep/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: PreparationProcedure

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PreparationProcedure		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PreparationProcedure**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PreparationProcedure/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SampleCollection

Conformance Class

http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleCollection		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleCollection**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleCollection/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: PhysicalDimension

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PhysicalDimension		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PhysicalDimension**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PhysicalDimension/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: NamedLocation

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-NamedLocation	
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-NamedLocation**
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-NamedLocation/test-1**

	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: StatisticalClassification

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-StatisticalClassification		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-StatisticalClassification**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-StatisticalClassification/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SampleType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleType		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleType**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleType/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SampleTypeByGeometryType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleTypeByGeometryType		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleTypeByGeometryType**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleTypeByGeometryType/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SamplerType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SamplerType		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SamplerType**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SamplerType/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

EF

Conformance class: MonitoringFacility

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacility	
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacility**

Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacility/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-geometryRequired

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacility-geometryRequired		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-geometryRequired**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-geometryRequired/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetwork

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetwork		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetwork/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.

	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringActivity

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivity		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivity**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivity/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgram

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgram		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgram/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

EF Extended

MonitoringFacilityExt

Conformance class: MonitoringFacilityExt-reportedTo

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-reportedTo		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-reportedTo**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-reportedTo/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-mediaMonitored

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-mediaMonitored		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-mediaMonitored**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-mediaMonitored/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-legalBackground

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-legalBackground		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-legalBackground**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-legalBackground/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-responsibleParty

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-responsibleParty		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-responsibleParty**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-responsibleParty/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-onlineResource

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-onlineResource	
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-onlineResource**

Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-onlineResource/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-purpose

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-purpose		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-purpose**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-purpose/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-measurementRegime

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-measurementRegime		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-measurementRegime**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-measurementRegime/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.

	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-mobile

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-mobile		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-mobile**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-mobile/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-resultAcquisitionSource

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-resultAcquisitionSource		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-resultAcquisitionSource**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-resultAcquisitionSource/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-specializedMFType

<table>
<tr><th colspan="3">Conformance Class</th></tr>
<tr><td colspan="3">http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-specializedMFType</td></tr>
<tr><td>Requirement</td><td colspan="2">http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-specializedMFType</td></tr>
<tr><td rowspan="3">Test</td><td colspan="2">http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-specializedMFType/test-1</td></tr>
<tr><td>Test purpose</td><td>Verify that all requirements from the requirements class have been fulfilled.</td></tr>
<tr><td>Test method</td><td>Inspect the documentation of the application, schema or profile.</td></tr>
</table>

Conformance class: MonitoringFacilityExt-supersedes

<table>
<tr><th colspan="3">Conformance Class</th></tr>
<tr><td colspan="3">http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-supersedes</td></tr>
<tr><td>Requirement</td><td colspan="2">http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-supersedes</td></tr>
<tr><td rowspan="3">Test</td><td colspan="2">http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-supersedes/test-1</td></tr>
<tr><td>Test purpose</td><td>Verify that all requirements from the requirements class have been fulfilled.</td></tr>
<tr><td>Test method</td><td>Inspect the documentation of the application, schema or profile.</td></tr>
</table>

Conformance class: MonitoringFacilityExt-supersededBy

<table>
<tr><th colspan="2">Conformance Class</th></tr>
<tr><td colspan="2">http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-supersededBy</td></tr>
<tr><td>Requirement</td><td>http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-supersededBy</td></tr>
</table>

Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-supersededBy/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

MonitoringNetwork

Conformance class: MonitoringNetworkExt-mediaMonitored

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-mediaMonitored		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-mediaMonitored**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-mediaMonitored/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-legalBackground

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-legalBackground	
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-legalBackground**
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-legalBackground/test-1**

	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-responsibleParty

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-responsibleParty		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-responsibleParty**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-responsibleParty/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-onlineResource

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-onlineResource		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-onlineResource**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-onlineResource/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-purpose

Conformance Class			
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-purpose			
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-purpose**		
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-purpose/test-1**		
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.	
	Test method	Inspect the documentation of the application, schema or profile.	

Conformance class: MonitoringNetworkExt-organizationLevel

Conformance Class			
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-organizationLevel			
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-organizationLevel**		
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-organizationLevel/test-1**		
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.	
	Test method	Inspect the documentation of the application, schema or profile.	

Conformance class: MonitoringNetworkExt-supersedes

Conformance Class

http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-supersedes		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-supersedes**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-supersedes/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-supersededBy

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-supersededBy		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-supersededBy**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-supersededBy/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

MonitoringActivity

Conformance class: MonitoringActivityExt-activityTime

Conformance Class
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-activityTime

Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-activityTime**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-activityTime/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringActivityExt-activityConditions

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-activityConditions		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-activityConditions**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-activityConditions/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringActivityExt-responsibleParty

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-responsibleParty	
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-responsibleParty**
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-responsibleParty/test-1**

	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringActivityExt-onlineResource

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-onlineResource		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-onlineResource**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-onlineResource/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

MonitoringProgram

Conformance class: MonitoringProgramExt-mediaMonitored

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-mediaMonitored		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-mediaMonitored**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-mediaMonitored/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.

	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgramExt-legalBackground

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-legalBackground		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-legalBackground**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-legalBackground/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgramExt-responsibleParty

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-responsibleParty		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-responsibleParty**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-responsibleParty/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgramExt-onlineResource

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-onlineResource		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-onlineResource**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-onlineResource/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgramExt-purpose

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-purpose		
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-purpose**	
Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-purpose/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgramExt-supersedes

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-supersedes	
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-supersedes**

Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-supersedes/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgramExt-supersededBy

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-supersededBy	
Requirement	**http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramExt-supersededBy**

Test	**http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramExt-supersededBy/test-1**	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.